

IoT Architecture & Cloud Platforms

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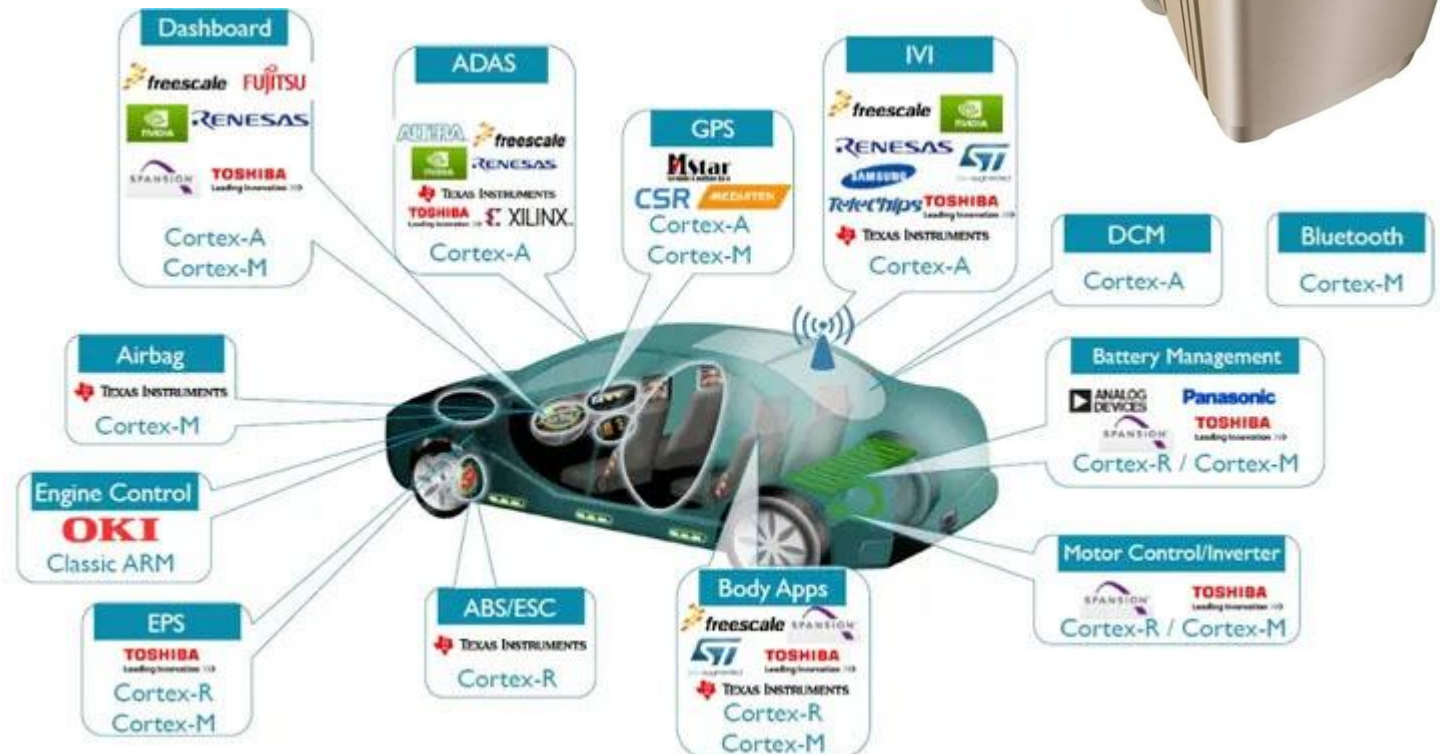
2026

What is an Embedded System?

- An **Embedded System** is a specialized computer system designed to perform a specific task or a set of tasks within a larger device. Unlike a general-purpose computer (such as a laptop), an embedded system is dedicated to controlling, monitoring, or operating a particular function.
- Characteristics: Designed for a specific purpose | Real-time operation (responds quickly to events) | Compact size | Low power consumption | Reliable and efficient.
- **Components of an Embedded System**
 - **Microcontroller/Microprocessor** – The brain of the system.
 - **Memory** – Stores programs and data.
 - **Input Devices** – Sensors, switches, keyboards, etc.
 - **Output Devices** – Displays, motors, LEDs, speakers, etc.
 - **Software/Firmware** – Program that controls the hardware.

Examples for Embedded Systems

- **Washing Machine** – Controls washing cycles.
- **Microwave Oven** – Manages cooking time and temperature.
- **Automobiles** – Engine control, airbags, ABS systems.
- **Smartwatches** – Monitor health and display notifications.
- **ATM Machines** – Handle banking transactions.



What is Physical Computing?

- **Physical Computing** is the design and creation of interactive systems that use hardware and software to sense and respond to the physical world. It bridges the gap between the digital world and the physical world.
 - E.g., A smart lamp that changes brightness based on ambient light.
- A physical computing system typically:
 - **Senses** the environment using sensors
 - **Processes** information using a microcontroller or computer
 - **Acts** on the environment using actuators
 - Input → Processing → Output

What is IoT?

- IoT = Physical Device + Connectivity + Data Exchange
- **Internet of Things (IoT)** refers to a network of physical devices ("things") that are equipped with sensors, actuators, processors, software, and communication capabilities, allowing them to collect data, exchange information, and interact with other devices or cloud services over a network (typically the Internet).

Feature	Physical Computing	IoT
Focus	Interaction with physical world	Connected smart systems
Internet Required	No	Usually Yes
Networking	Optional	Essential
Data Sharing	Local	Remote
Cloud Integration	Not required	Common
Mobile Access	Usually absent	Common
Scale	Single device	Multiple connected devices
Analytics	Local processing	Cloud analytics possible

Comparison

Feature	Embedded System	Physical Computing	IoT
Dedicated computing	✓	✓	✓
Sensors	Optional	✓	✓
Actuators	Optional	✓	✓
Human interaction	Limited	Extensive	Extensive
Internet connection	Usually no	Optional	Required
Cloud services	No	Optional	Yes
Remote monitoring	No	Usually no	Yes
Device-to-device communication	Rare	Optional	Common

Device	Embedded System	Physical Computing	IoT
Digital calculator	✓	✗	✗
Microwave oven	✓	Limited	✗
Automatic hand dryer	✓	✓	✗
Line-following robot	✓	✓	✗
Smart irrigation system	✓	✓	✓
Smart electricity meter	✓	✓	✓
Smart weather station	✓	✓	✓
Smart door lock	✓	✓	✓

What is a Cyber-Physical System (CPS)?

- U.S. NSF: A Cyber-Physical System is a system in which computational and physical components are deeply integrated, with each operating on different spatial and temporal scales, exhibiting multiple and distinct behavioral modalities, and interacting with each other in ways that change with context.
- Simply stated: A Cyber-Physical System is an intelligent system that continuously senses, computes, communicates, and controls physical processes in real time.
 - E.g., Smart Car - The system continuously monitors traffic and adjusts vehicle behavior in real time. Smart Factory - Everything works together through continuous feedback. Smart Power Grid. Autonomous Drone
- A CPS forms a **closed feedback loop**: Sense the environment. Process the data. Make decisions. Control the physical system. Repeat continuously.

What is Edge Computing?

- **Edge Computing** is a distributed computing paradigm in which data processing and decision-making are performed **at or near the source of data generation (the "edge")**, rather than sending all data to a centralized cloud server. The "edge" refers to devices such as sensors, microcontrollers, gateways, smartphones, industrial controllers, or local servers that are close to where the data is produced.

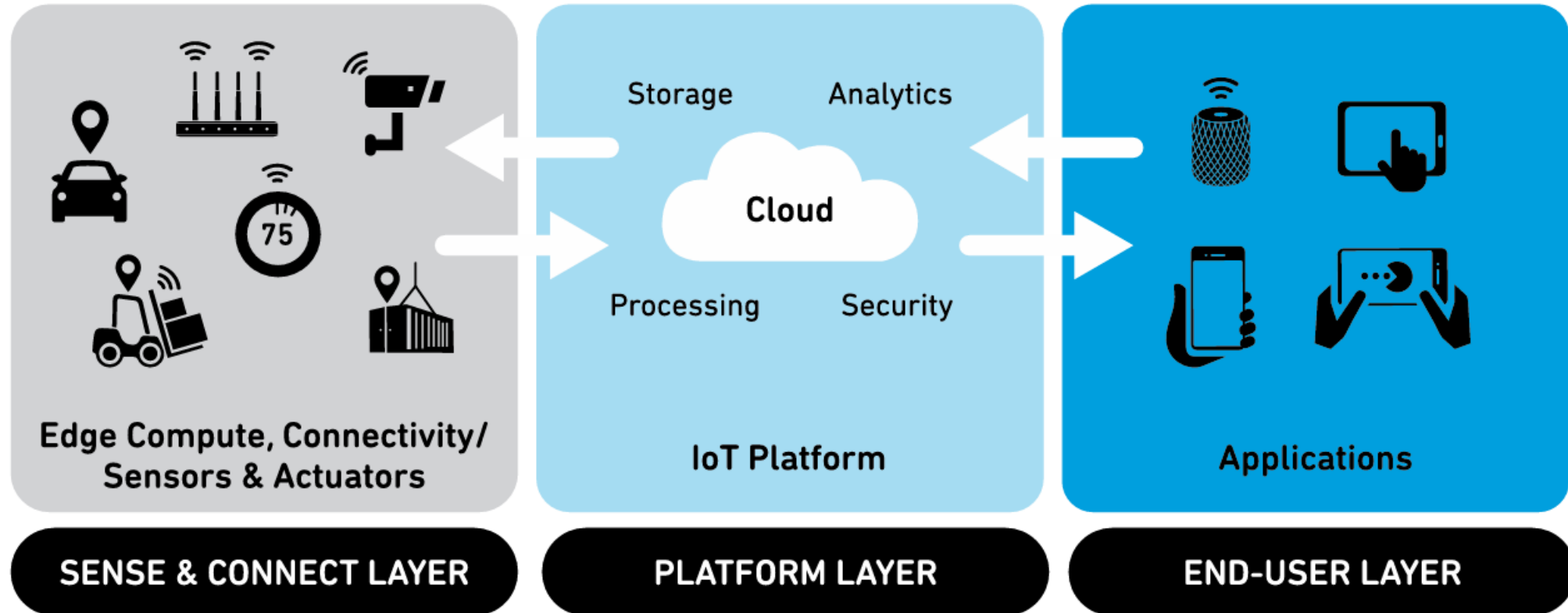
Summary

- **Embedded Systems** provide the computing platform.
- **Physical Computing** enables interaction with the physical environment.
- **Cyber-Physical Systems** integrate sensing, communication, and real-time feedback control.
- **IoT** connects these systems to networks and cloud services.
- **Edge Computing** optimizes where computation occurs by processing data near its source.

Aspect	Embedded System	Physical Computing	CPS	IoT	Edge Computing
Dedicated computer	✓	✓	✓	✓	✓
Sensors	Optional	✓	✓	✓	✓
Actuators	Optional	✓	✓	✓	✓
Physical interaction	Limited	✓	✓	✓	✓
Continuous feedback control	Limited	Basic	✓	✓	✓
Internet connectivity	No	Optional	Optional	✓	Optional
Cloud services	No	Optional	Optional	✓	Optional
Local data processing	✓	✓	✓	✓	✓ (primary focus)
Remote monitoring	No	Optional	Optional	✓	Optional
Primary goal	Device control	Physical interaction	Intelligent control	Connected systems	Fast, local processing

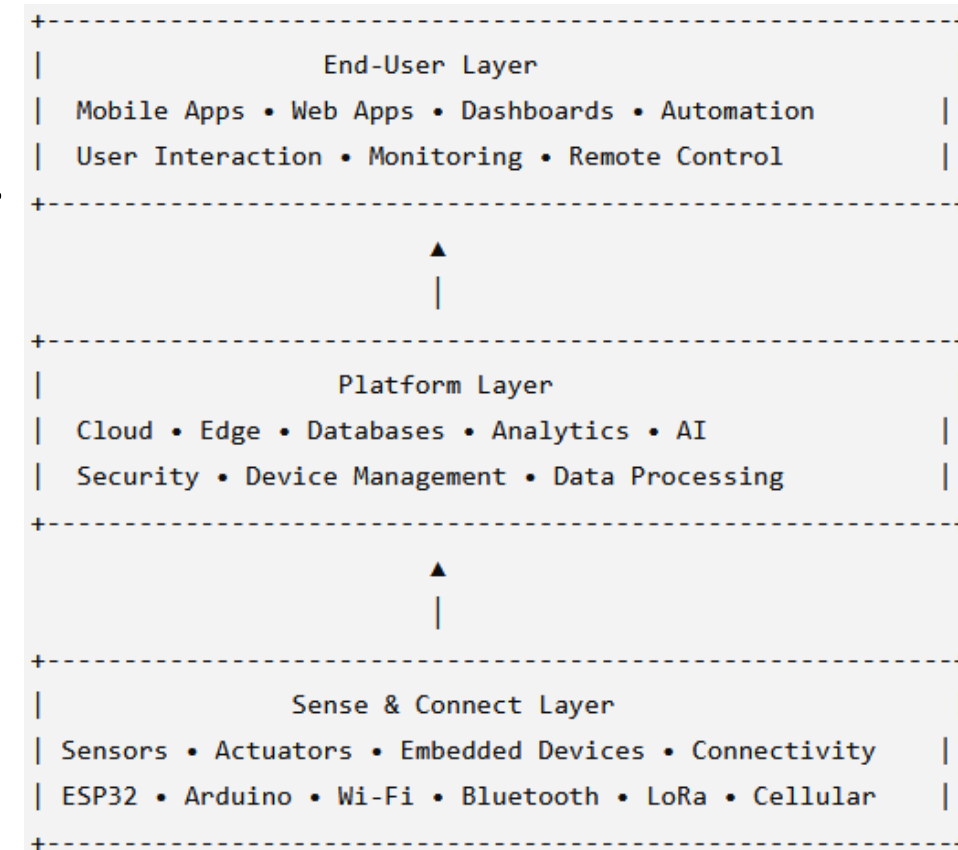
IoT Architecture

Simplified Three Layer Architecture (Qorvo)



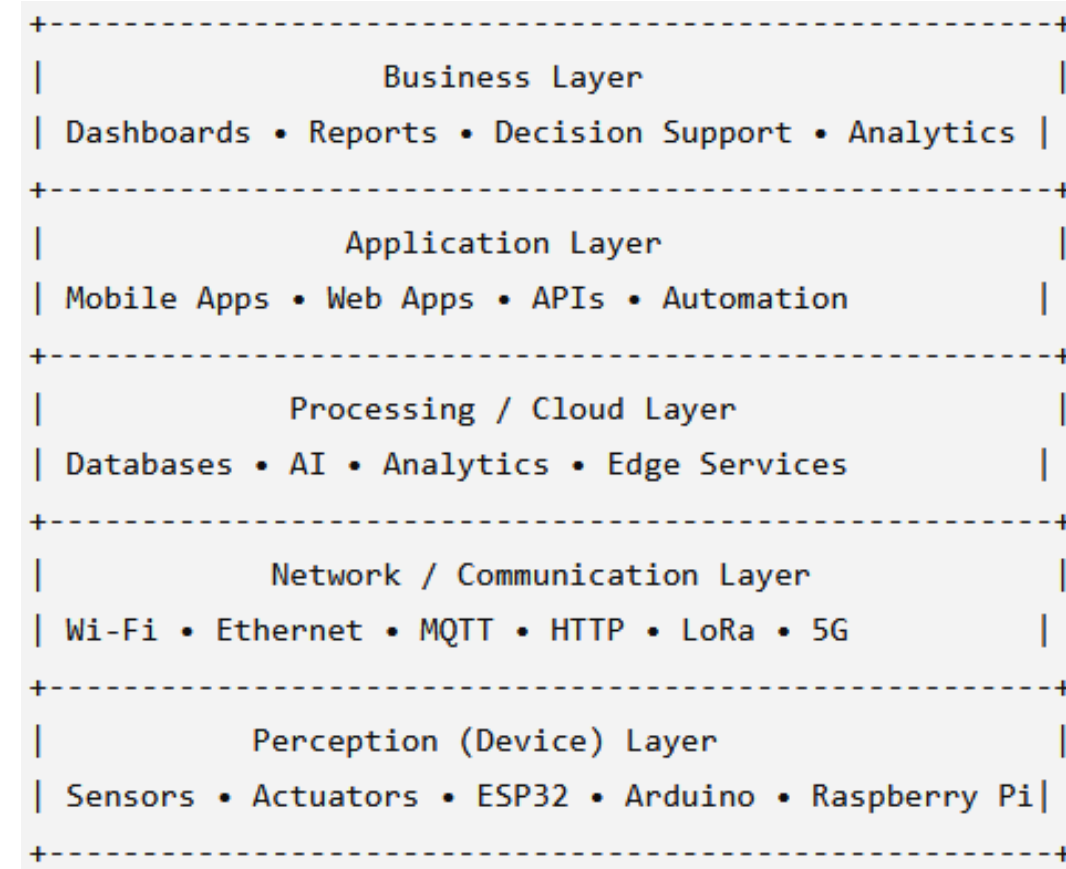
Simplified Three Layer Architecture Cont.

- The **Sense & Connect layer** is the foundation of every IoT system. It includes all physical devices that interact with the real world and the communication technologies that connect them to higher-level services.
- The **Platform layer** is the "brain" of the IoT system. It receives data from devices (can be thousands), manages those devices, and transforms raw measurements into meaningful information.
 - Rather than being limited to the cloud, modern IoT platforms may also include **edge computing**, where some processing occurs close to the devices before data reaches the cloud.
- The **End-User layer** is where people interact with the IoT system. It provides applications that present processed information and allow users to control connected devices.



The Five Layer Architecture

- The **perception (device) layer** is responsible for sensing and interacting with the physical world.
- The **network (communication) layer** transfers data between devices, gateways, cloud platforms, and applications.
- The **processing (middleware/cloud/edge) layer** processes, stores, and analyzes data.
- The **application layer** provides services to users.
- The **business layer** supports organizational decision-making.



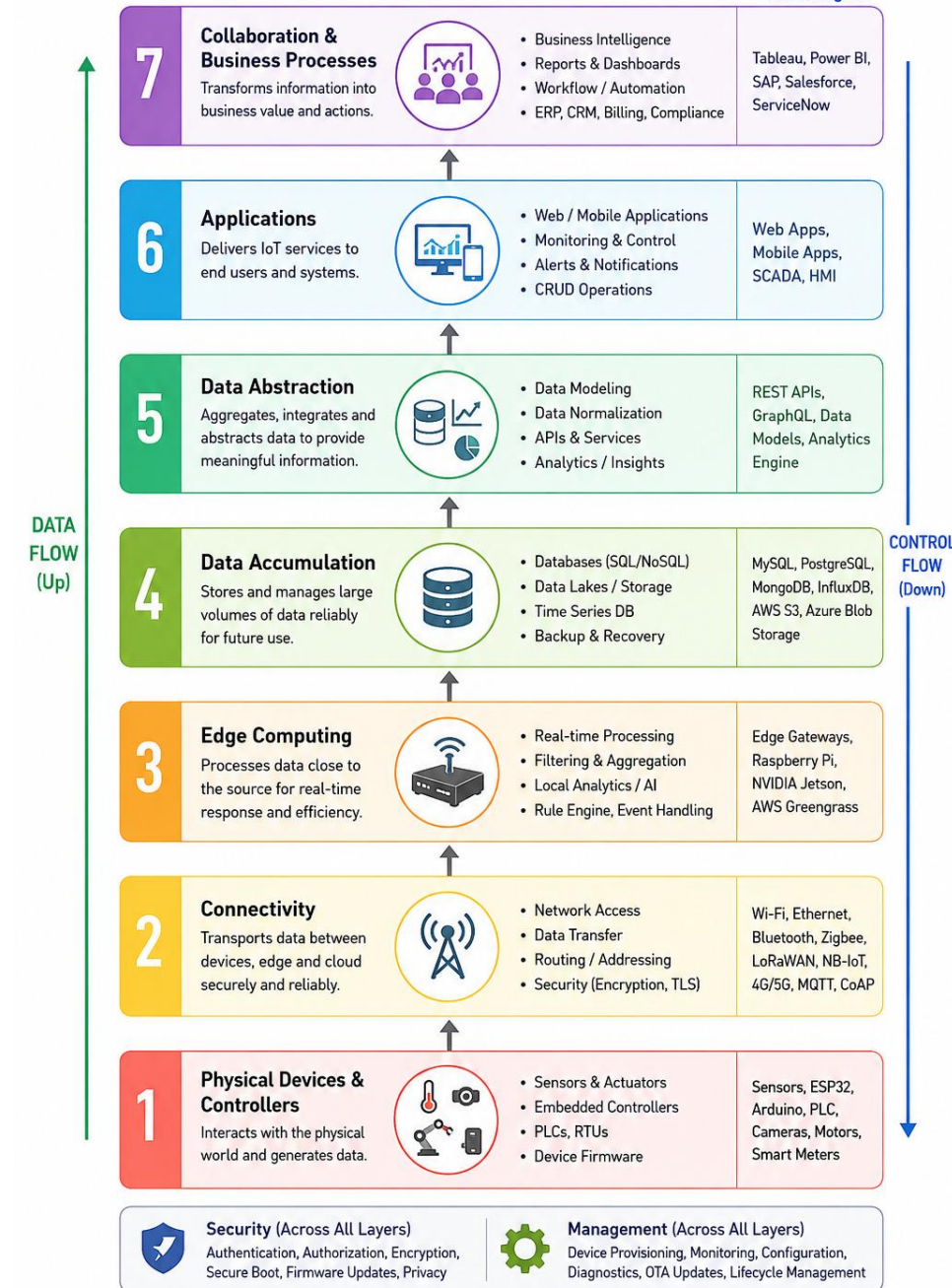
The Seven Layer Architecture Industrial IoT (IIOT)

- Unlike the simpler 3-layer or 5-layer models, the seven-layer model emphasizes the **flow of data** from physical devices to business decisions and the **flow of control commands** back to the devices.
- It is widely associated with the **Cisco Systems IoT Reference Model** and is particularly useful for understanding industrial IoT (IIoT) and large-scale enterprise IoT systems.
- Layers:
 - **Physical Devices & Controllers** – Sense and control the environment.
 - **Connectivity** – Transfer data between devices and systems.
 - **Edge Computing** – Perform local processing and real-time control.
 - **Data Accumulation** – Persist streaming data in storage.
 - **Data Abstraction** – Transform raw data into structured, meaningful information.
 - **Applications** – Deliver monitoring, visualization, and control to users.
 - **Collaboration & Business Processes** – Convert information into operational and business value.

7-LAYER IoT REFERENCE MODEL

Data flows up, while control and commands flow down.

Example Technologies



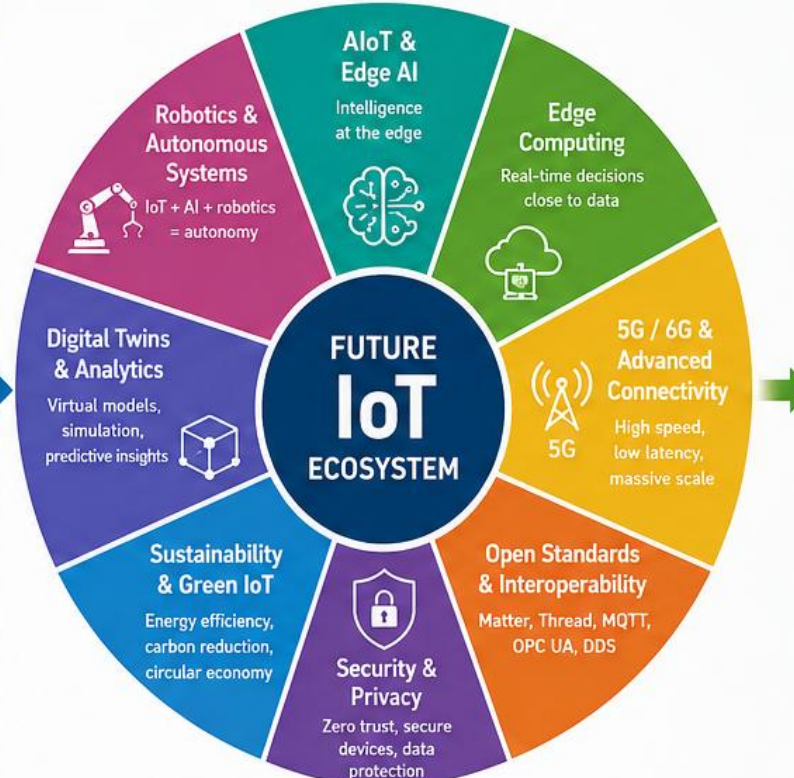
IoT MARKET TRENDS & PREDICTIONS – 2026 AND BEYOND

Intelligent • Connected • Autonomous • Secure • Sustainable

MAJOR IoT MARKETS



Strongest Growth Drivers: Digital transformation • Efficiency • Safety Sustainability • Customer experience

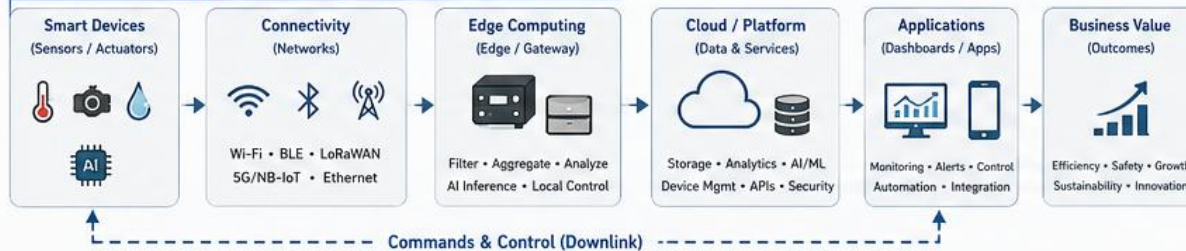


Outcome: Smarter operations • New business models • Better decisions • Higher efficiency • Better world

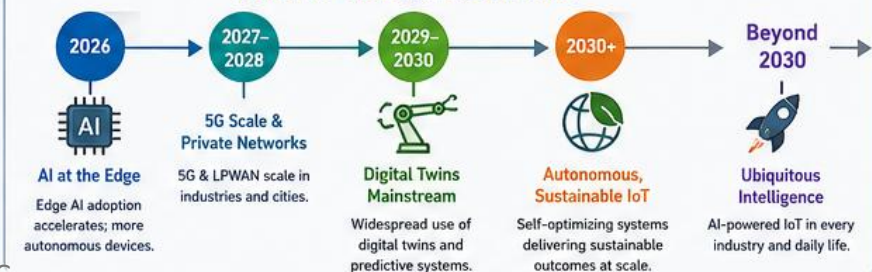
KEY TECHNOLOGY TRENDS (2026+)



EVOLVING IoT ARCHITECTURE



ROADMAP TO 2030 AND BEYOND



IoT Market Segments

Market	Typical Applications	Example Devices
Smart Home	Home automation, security, lighting	Smart thermostats, smart locks, smart speakers
Smart Buildings	Energy management, HVAC, access control	Smart meters, occupancy sensors
Industrial IoT (IIoT)	Factory automation, predictive maintenance	PLCs, industrial sensors, robots
Smart Cities	Traffic, parking, waste management	Smart streetlights, traffic sensors
Smart Healthcare	Remote patient monitoring, wearables	Smartwatches, glucose monitors
Smart Agriculture	Precision farming, livestock monitoring	Soil moisture sensors, weather stations
Smart Energy	Smart grids, renewable energy management	Smart meters, grid sensors
Transportation & Logistics	Fleet tracking, asset monitoring	GPS trackers, cold-chain sensors

Smart Home

- Smart homes connect everyday appliances to improve convenience, security, and energy efficiency.
- Typical applications: Smart lighting, Smart thermostats, Home security, Voice assistants, Smart appliances, Remote monitoring
- Key technologies: Wi-Fi, Bluetooth Low Energy (BLE), Zigbee, Thread, Matter
- **Current trend:** Interoperability through the **Matter** standard is reducing vendor lock-in and making devices from different manufacturers work together more seamlessly.



Industrial IoT (IIoT)

- Industrial IoT is one of the fastest-growing IoT sectors because it directly improves productivity and reduces operational costs.
- Applications: Predictive maintenance, Machine monitoring, Quality inspection, Digital twins, Warehouse automation, Asset tracking
- Benefits: Reduced downtime, Improved productivity, Better equipment utilization, Increased safety
- Industry analysts consistently identify IIoT as one of the largest and fastest-growing IoT application areas.



Smart Healthcare

- Healthcare is increasingly adopting IoT for continuous monitoring and personalized care.
- Applications: Wearable fitness devices, Remote patient monitoring, Smart hospital equipment, Medication tracking, Elderly care
- Benefits: Early disease detection, Continuous health monitoring, Reduced hospital visits, Improved patient outcomes



Major IoT Technology Trends

- **Edge Computing:**

- Processing is moving closer to devices.
- Advantages: Lower latency, Reduced bandwidth, Faster decision making, Better privacy
- Edge computing is now considered a fundamental architectural element of modern IoT systems.

- **Artificial Intelligence (AI):**

- Rather than relying solely on cloud AI, many systems now perform **AI inference at the edge**, enabling real-time decisions.

- **5G and Advanced Wireless Connectivity:**

- High-speed, low-latency wireless communication enables: Autonomous vehicles, Smart factories, Remote robotics, Massive IoT deployments
- Other important technologies include: Wi-Fi 6/7, Bluetooth LE, Thread, Zigbee, LoRaWAN, NB-IoT

Major IoT Technology Trends Cont.

- **Matter Standard:**

- One of the biggest developments in consumer IoT is the adoption of **Matter**, which provides interoperability among smart home devices from different manufacturers.
- Benefits: Easier installation, Cross-vendor compatibility, Improved user experience

- **Cybersecurity:**

- As billions of devices connect to networks, security has become a top priority.
- Key requirements: Secure boot, Device authentication, End-to-end encryption, Firmware updates, Identity management

- **Digital Twins:**

- A digital twin is a virtual representation of a physical asset that is continuously updated using IoT data.

- **Sustainability:**

- Sustainability is a major driver for smart buildings, smart grids, and precision agriculture.

Wireless Standards that Enable the Internet of Things (IoT)

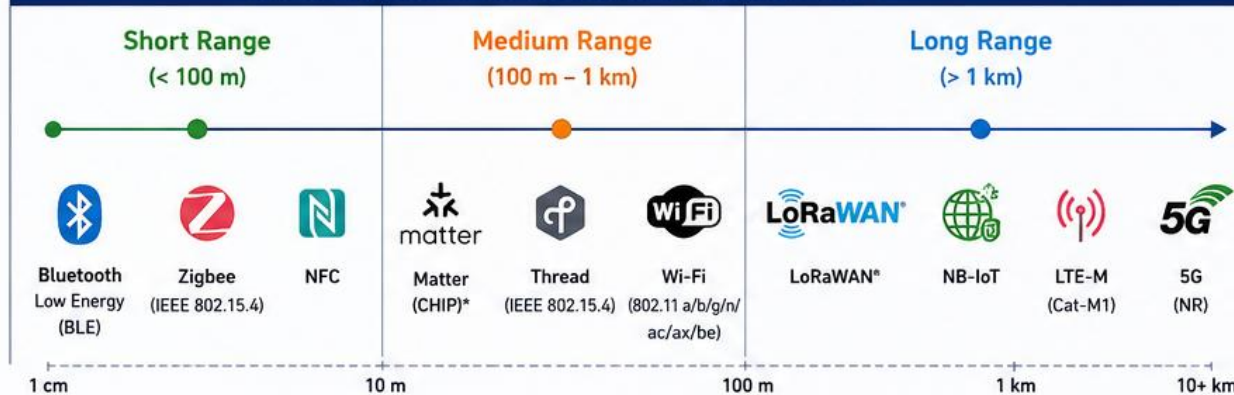
IoT Wireless Standards

- Wireless communication is one of the key technologies that enables the Internet of Things. Different IoT applications have different requirements in terms of **range, data rate, power consumption, cost, and network size**, so no single wireless technology is suitable for every application.
- Key Selection Criteria:
 - Communication range
 - Data rate (bandwidth)
 - Power consumption
 - Network topology
 - Device cost
 - Number of connected devices
 - Internet connectivity requirements

IoT WIRELESS TECHNOLOGIES

Classification and Technologies by Communication Range, Data Rate, Power and Use Cases

CLASSIFICATION BY COMMUNICATION RANGE



* Matter (formerly CHIP) is an application-layer interoperability standard that runs over IP networks such as Thread, Wi-Fi and Ethernet.

KEY FACTORS FOR TECHNOLOGY SELECTION

- Range / Coverage** – How far the signal can reach
- Data Rate (Bandwidth)** – How much data can be transmitted
- Power Consumption** – Impact on battery life
- Network Topology** – Star, Mesh, Cellular, etc.
- Cost** – Module and infrastructure cost
- Scalability** – Number of devices supported
- Security** – Encryption, authentication, privacy
- Internet Connectivity** – Direct IP or via gateway

COMPARISON OF MAJOR IoT WIRELESS TECHNOLOGIES

Technology	NFC	Bluetooth LE	Zigbee (802.15.4)	Thread (802.15.4)	Matter (CHIP)* (over Thread/ Wi-Fi/Ethernet)	Wi-Fi (802.11)	LoRaWAN®	NB-IoT	LTE-M (Cat-M1)	5G (NR)
Typical Range	< 10 cm	10 – 100+ m	10 – 100 m (per hop)	10 – 100 m (per hop)	Depends on underlying technology	30 – 100 m (indoor)	2 – 15+ km	Several km	Several km	Several km
Data Rate	106 – 424 kbps	125 kbps – 2 Mbps	250 kbps	250 kbps	Depends on underlying technology	Mbps – Gbps	0.3 – 50 kbps	~20 – 250 kbps	Up to ~1 Mbps	10s Mbps – Gbps
Power Consumption	Very Low	Very Low	Very Low	Very Low	Depends on underlying technology	High	Very Low	Low	Low	Medium
Topology	Point-to-point	Star	Mesh (Star)	Mesh	IP-based (uses Thread/ Wi-Fi/Ethernet)	Star	Star-of-stars	Cellular	Cellular	Cellular
Internet Connectivity	No (P2P)	Through gateway	Through gateway	Native IP (via Border Router)	Native IP	Native IP	Through gateway (Network Server)	Native IP (Cellular)	Native IP (Cellular)	Native IP (Cellular)
Typical Use Cases	Payments, access control, pairing	Wearables, beacons, health devices	Home automation, lighting, sensors	Smart home, sensors, control	Smart home device, interoperability	Cameras, gateways, high data apps	Agriculture, smart cities, tracking	Smart meters, asset tracking, sensors	Wearables, fleet management	Industrial IoT, autonomous systems, robots

COMMON IoT APPLICATION DOMAINS



LEGEND

- Best for long range
- Best for high data rate
- Lower cost
- Best for low power
- Strong security

Key Takeaway

Choose the right wireless technology based on the required range, data rate, power, cost and application needs.

Near Field Communication (NFC)

Short Range • High Security • Easy to Use • Low Power

WHAT IS NFC?

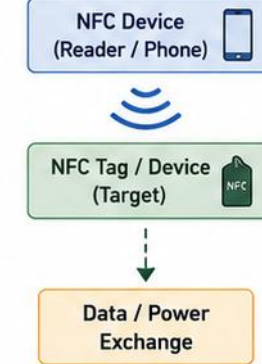
NFC is a very short-range wireless technology that enables two devices to communicate when they are brought very close together (typically less than 10 cm).

-  Operates at 13.56 MHz
-  Very low power
-  Secure due to proximity
-  Simple pairing and use



HOW NFC WORKS

- 1 When two NFC devices come within a few centimeters, they detect each other.
- 2 An RF field is created (13.56 MHz) to establish a communication link.
- 3 Data is exchanged using simple, secure protocols.
- 4 The connection closes automatically when devices are separated.



KEY CHARACTERISTICS

- ✓ Range: < 10 cm
- ✓ Frequency: 13.56 MHz
- ✓ Data Rate: Up to 424 kbps
- ✓ Power: Very Low
- ✓ Topology: Point-to-point
- ✓ Standards: ISO/IEC 14443, ISO/IEC 15693, NFC Forum

SUITABILITY FOR IoT

Why NFC is Well Suited for IoT	Benefits for IoT Applications
 Very short range (< 10 cm)	Reduces risk of eavesdropping, increases security and user control.
 Very low power consumption	Ideal for battery-powered sensors, tags, and wearables.
 Simple and fast interaction	Tap to connect, pair, configure, or exchange data instantly.
 Secure by design	Physical proximity + encryption prevents unauthorized access.
 Low cost and widely supported	NFC chips and tags are inexpensive and supported in most smartphones.
 Works without internet	Enables local device setup, data transfer, and control offline.

COMMON IoT USE CASES

 Device Pairing & Provisioning Tap to quickly pair or configure IoT devices.	 Access Control & Security NFC cards, badges or smart locks.	 Identification & Authentication Verify users or devices in IoT systems.	 Asset Tracking & Tagging NFC tags on assets for identification.
 Contactless Payments & Transactions Secure tap-and-pay applications.	 Information Access (NFC Posters) Tap to get product info, URLs, manuals, etc.	 Device Configuration & Maintenance Tap to read/write settings or firmware.	 Healthcare & Wearables Patient ID, health data access, or device configuration.



NFC brings secure, seamless, and low-power connectivity to IoT devices, enabling effortless interaction between people and smart objects.

Miran Ekanayake @ UCSC

Applications: Smart Home • Industrial IoT • Healthcare • Retail • Logistics • Smart Cities



BLUETOOTH LOW ENERGY (BLE) AND ITS SUITABILITY FOR IoT

Low power • Short range • Cost effective • Easy connectivity • Ideal for IoT devices

WHAT IS BLE?

Bluetooth Low Energy (BLE) is a low power wireless communication technology designed for short-range data exchange between devices.

- Part of the Bluetooth 5.x / 6.0 standard (IEEE 802.15.1)
- Optimized for very low power consumption
- Supports connectionless (advertising) and connection-oriented communication
- Operates in 2.4 GHz ISM band

BLE NETWORK TOPOLOGY

PERIPHERAL DEVICES (Sensors / Actuators)



CENTRAL DEVICE (Observer / Controller)



Smartphone
/ Tablet / PC
/ IoT Hub

CLOUD / APPLICATION



Cloud Server



Dashboard /
Analytics



Mobile App
(User)



Internet

---> BLE Link (Short Range) <--> IP / Wi-Fi / Cellular

WHY BLE IS SUITABLE FOR IoT



Very Low Power
Designed for coin-cell or small battery operation (months to years).



Short Range
Typically 10 – 100 meters, reduces interference and improves security.



Low Cost
Simple, low-cost chips and minimal infrastructure.



Ubiquitous Support
Built into almost all smartphones, tablets and many other devices.



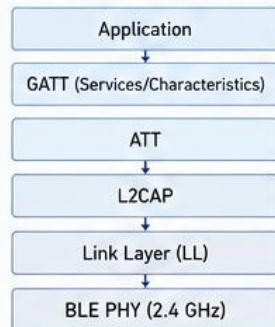
Flexible Topologies
Supports star, mesh and broadcast (advertising) modes.



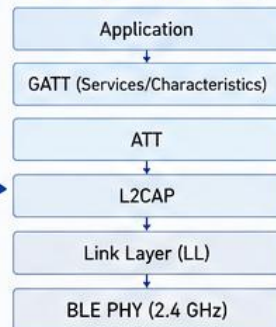
Secure
AES-CCM encryption and privacy features ensure secure data.

BLE ARCHITECTURE (Simplified)

Peripheral (Sensor)



Central (Smartphone / Hub)



BLE COMMUNICATION MODES



1. Connected Mode

Central and peripheral devices establish a connection for reliable, two-way data exchange.



2. Advertising / Broadcast Mode

Peripheral periodically sends advertisements. Useful for beacons, location and discovery.



3. Mesh Networking (BLE Mesh)

Many-to-many device communication for large IoT deployments (lighting, sensors, industrial).

TYPICAL IoT APPLICATIONS USING BLE



Wearables



Smart Home
(locks, lights)



Environmental
Monitoring



Asset
Tracking



Beacons & Indoor
Positioning



Medical &
Healthcare



Automotive
(Access, TPMS)



Industrial
Sensors

BLE AT A GLANCE



Range
10 – 100 m
(Class 2 typical)



Data Rate
Up to 2 Mbps
(BLE 5.x)



Power
Ultra low power
(mA to μ A)



Frequency
2.402 – 2.480 GHz
(40 channels)



Topology
Star, Mesh,
Broadcast



Cost
Low
(BOM friendly)

Wi-Fi (IEEE 802.11) for IoT

High-speed, widely available wireless connectivity for Internet of Things



WHAT IS WI-FI?

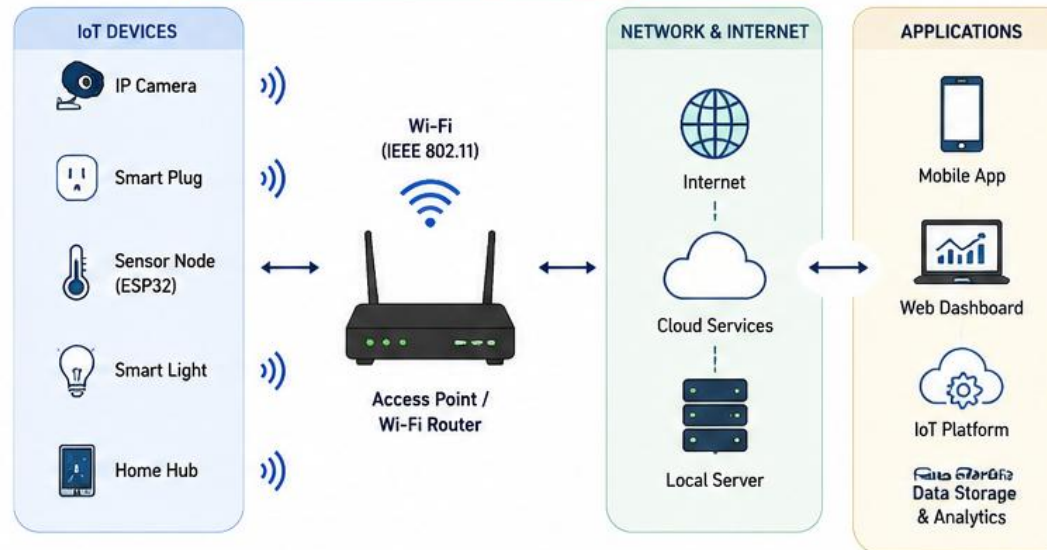
Wi-Fi (IEEE 802.11) is a family of wireless networking standards that provide local area network (LAN) connectivity using radio waves.

Devices connect to a Wi-Fi Access Point (AP) which provides access to the local network and the Internet.

KEY FEATURES

- High Data Rate**
From tens of Mbps to several Gbps (Wi-Fi 4/5/6/6E/7)
- Internet Connectivity**
Direct connection to IP networks and cloud services
- Existing Infrastructure**
Leverages widely available Wi-Fi routers and access points
- Strong Security**
WPA2/WPA3 encryption and authentication
- Many Devices**
Supports multiple simultaneous devices

WI-FI IoT ARCHITECTURE



Wi-Fi GENERATIONS AT A GLANCE

Standard	Release	Bands	Max Data Rate (Theoretical)	Highlights
802.11n (Wi-Fi 4)	2009	2.4 / 5 GHz	Up to 600 Mbps	MIMO, better range and speed
802.11ac (Wi-Fi 5)	2014	5 GHz	Up to 6.9 Gbps	Higher throughput
802.11ax (Wi-Fi 6/6E)	2019 / 2021	2.4 / 5 / 6 GHz	Up to 9.6 Gbps	OFDMA, MU-MIMO, better efficiency
802.11be (Wi-Fi 7)	2024+	2.4 / 5 / 6 GHz	Up to 46 Gbps+	Multi-Link Operation, very low latency

SUITABILITY FOR IoT

CRITERIA	SUITABILITY
Data Rate / Bandwidth	Very High ✓✓✓
Range	Short-Medium (30-100 m indoor) ✓
Power Consumption	Higher than LPWAN and BLE ✗
Latency	Low ✓✓
Network Topology	Infrastructure (Star) ✓
Number of Devices	High (dozens to hundreds per AP) ✓
Internet Connectivity	Native IP connectivity ✓✓✓
Security	Strong (WPA2/WPA3) ✓✓
Module / Device Cost	Low-Medium ✓
Battery Life	Hours to days (less for high data) ✗
Best For	High data rate, frequent communication, mains powered devices ✓✓✓

TYPICAL IoT USE CASES USING WI-FI



CONSIDERATIONS / LIMITATIONS

- ✗ Higher power consumption – not ideal for long battery life
- ✗ Range limited indoors – affected by walls and interference
- ✗ More expensive than BLE/Zigbee/Thread
- ✗ Requires infrastructure (AP/router)
- ✗ Spectrum congestion in 2.4 GHz band possible

BEST PRACTICES FOR IoT OVER WI-FI

- ✓ Use WPA3 for security
- ✓ Use 2.4 GHz for range, 5/6 GHz for speed
- ✓ Enable power-save/DTIM for battery devices
- ✓ Use MQTT/HTTP for efficient communication
- ✓ Keep firmware up to date (OTA updates)

Wi-Fi is ideal for IoT devices that need high data rates, direct Internet access, and integration with cloud services, especially where power is not a strict constraint (mains powered or frequently charged devices).



Examples: ESP32 / ESP32-C3, Raspberry Pi, IP Cameras, Smart Home Hubs, Multimedia Devices, Industrial Gateways

ZIGBEE (IEEE 802.15.4) – IDEAL WIRELESS FOR IoT

Low Power • Low Cost • Reliable • Mesh Networking

WHAT IS ZIGBEE?

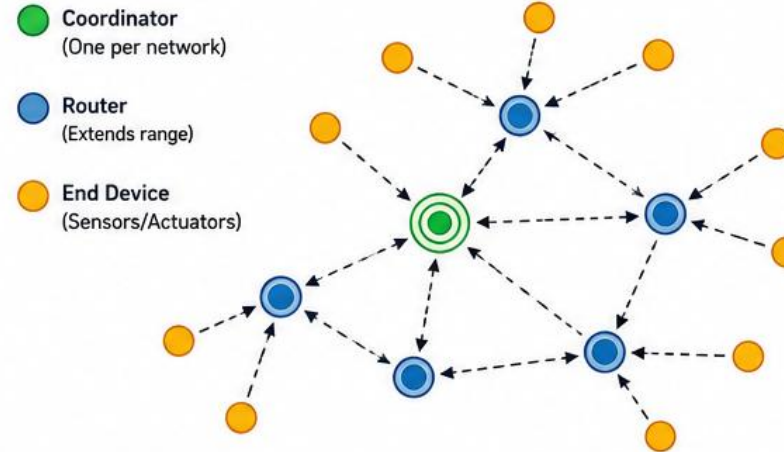
Zigbee is a low-power, low-data rate wireless communication standard based on IEEE 802.15.4 for short-range IoT applications.



KEY FEATURES

- ✓ Low Power – months to years on batteries
- ✓ Low Cost – simple, low-cost chipsets
- ✓ Mesh Networking – self-healing & reliable
- ✓ Scalable – supports thousands of devices
- ✓ Secure – AES-128 encryption
- ✓ Operates in unlicensed spectrum worldwide

ZIGBEE NETWORK TOPOLOGY (MESH)



Data can take multiple hops through routers to reach the coordinator.
If one path fails, the network self-heals automatically.

WHY ZIGBEE IS SUITABLE FOR IoT

Very Low Power	Designed for long battery life. Ideal for sensors and actuators.
Mesh Networking	Extends coverage, increases reliability, self-healing.
Low Cost	Inexpensive chips and simple implementation.
Short Range, Low Data Rate	Perfect for monitoring and control (10–100 m per hop, 250 kbps).
Secure	AES-128 encryption with authentication and access control.
Large Network Size	Supports up to 65,000+ devices in a single network.

ZIGBEE PROTOCOL STACK

Application	Application Framework (ZDO)
	Application Objects
Network	Network Layer (NWK)
	Routing, Network Security, Device Mgmt.
MAC	MAC Layer (IEEE 802.15.4 MAC) Channel Access, ACK, Security
PHY	PHY Layer (IEEE 802.15.4 PHY) 868/915 MHz / 2.4 GHz Radio

Data Rate
250 kbps

Frequency Bands
868/915 MHz
2.4 GHz

Range
10 – 100 m
(per hop)

ZIGBEE DEVICE TYPES



Coordinator (FFD)
Creates and manages the network. One per network. Usually mains powered.



Router (FFD)
Forwards data, extends range, maintains routes. Mains or battery powered.



End Device (RFD or FFD)
Sends/receives data to parent router. Very low power, battery operated.



APPLICATION EXAMPLES



Smart Lighting



Home Automation



Security Systems



Smart Energy
(Meters, plugs)



Industrial Monitoring



Environmental
Monitoring



Healthcare
Devices



Smart Agriculture

SUMMARY

Zigbee (IEEE 802.15.4) is a proven wireless standard for IoT applications that need low power, low cost, and reliable mesh networking. It is widely used in smart homes, industrial automation, smart energy, and many other IoT solutions.



Low Power



Low Cost



Reliable Mesh



Secure



Scalable



THREAD: A MODERN WIRELESS NETWORK FOR IoT

Thread is a low-power, secure, IPv6-based mesh networking protocol designed for IoT devices and smart home applications.

WHAT IS THREAD?

- IPv6-based mesh networking protocol (IEEE 802.15.4)
- Designed for low power, reliability and scalability
- Self-healing mesh network
- Supports thousands of devices
- Native IP connectivity to the Internet
- Secure by design (DTLS, AES-128)
- Runs on 2.4 GHz IEEE 802.15.4 radio

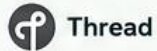
THREAD STACK (Simplified)

Application (CoAP, MQTT, ...)

UDP / DTLS

IPv6

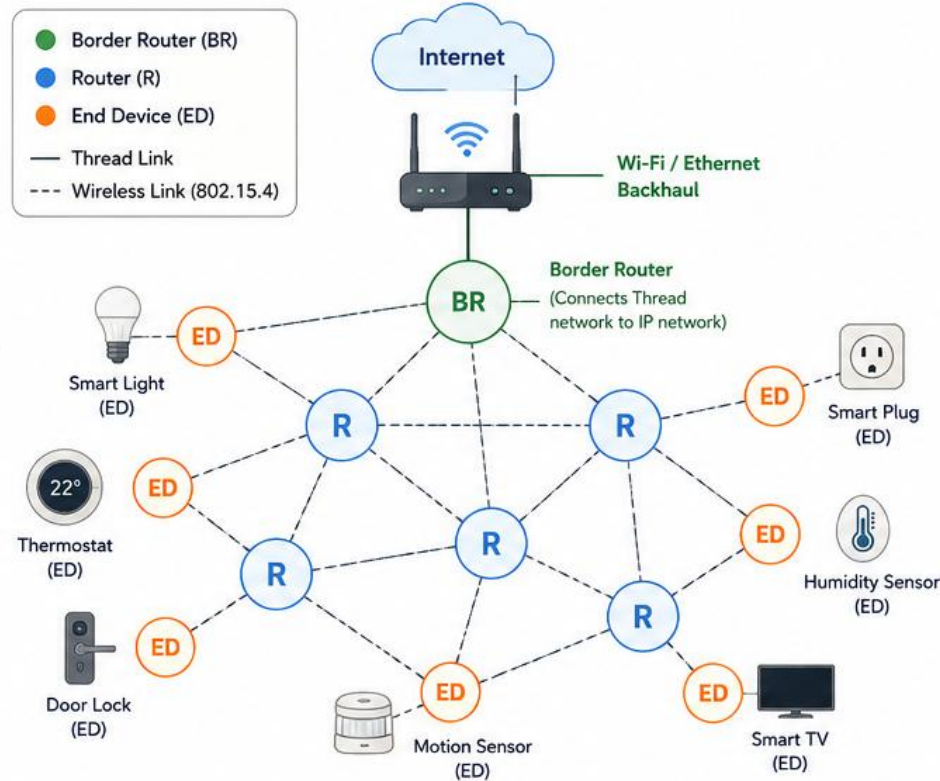
6LoWPAN



IEEE 802.15.4 MAC

IEEE 802.15.4 PHY (2.4 GHz)

THREAD NETWORK TOPOLOGY



WHY THREAD IS SUITABLE FOR IoT

Requirement	How Thread Addresses It
Low Power Operation	Optimized for battery-powered devices; supports sleepy end devices for long battery life.
Reliability	Mesh network with self-healing and automatic rerouting ensures robust connectivity.
Scalability	Supports hundreds to thousands of devices in one network.
Security	Built-in security: AES-128 link encryption, DTLS, device attestation.
IP Connectivity	Native IPv6 connectivity enables direct communication with cloud services and the Internet.
Interoperability	Open standard (Connectivity Standards Alliance) and foundation for Matter.
Low Cost	Uses IEEE 802.15.4 (low-cost radio); ideal for cost-sensitive IoT devices.
Smart Home Focus	Designed for home automation, works seamlessly with Matter over Thread.

HOW THREAD WORKS



TYPICAL IoT APPLICATIONS



Thread, combined with IP, IPv6 and open standards like Matter, enables a future where billions of IoT devices can communicate securely, reliably and efficiently in smart homes, buildings and industries.

Hiran Ekanayake @ UCSC



Matter (CHIP) – The IoT Interoperability Standard

One protocol. Many devices. Any ecosystem.

Matter (formerly CHIP – Connected Home over IP) is an application-layer standard that enables smart home devices from different brands to work together securely over IP networks.

What is Matter (CHIP)?

- Interoperability**
Devices from different manufacturers work together seamlessly.
- Secure by Design**
Strong encryption, device attestation and secure commissioning.
- Local Control**
Works on local network without relying on cloud.
- Future Proof**
Built on open standards (IP) and designed for long-term evolution.

Matter Architecture

Matter runs over IP-based networks and supports multiple underlying technologies.

MATTER CONTROLLERS



Application Layer (Matter)

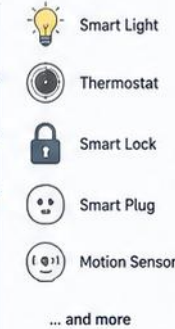


Interaction Model (Standardized Data Models & Commands)
e.g., On/Off, Brightness, Temperature, Lock, Motion, Energy, etc.

Security Layer

Authentication, Access Control, Encryption, Integrity

MATTER DEVICES



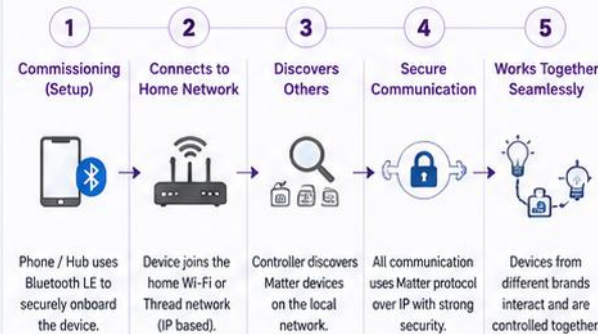
UNDERLYING NETWORKS (TRANSPORT LAYER)



Why Matter is Highly Suitable for IoT

	Interoperability	Breaks vendor lock-in. Devices from different brands work together on the same platform.
	Open & Standardized	Built on IP and open standards, ensuring wide adoption and long-term sustainability.
	Secure	End-to-end encryption, device authentication and secure onboarding protect the entire ecosystem.
	Local & Reliable	Local control keeps devices working even when the internet is down.
	Scalable	Supports thousands of devices in a home or enterprise environment.
	Easy for Developers	Standard data models and SDKs simplify development and reduce time-to-market.
	Future Ready	Designed to evolve with new device types, features and technologies.

How Matter Devices Work (Example)



Typical IoT Applications



Bottom Line

Matter (CHIP) brings the missing piece to the IoT puzzle – true interoperability. It allows devices to connect, communicate and cooperate securely, opening the door to a simpler, smarter and more connected world.

Matter + IP + Secure Networks
= The Foundation for the Future of IoT

Hiran Ekanayake @ UCSC

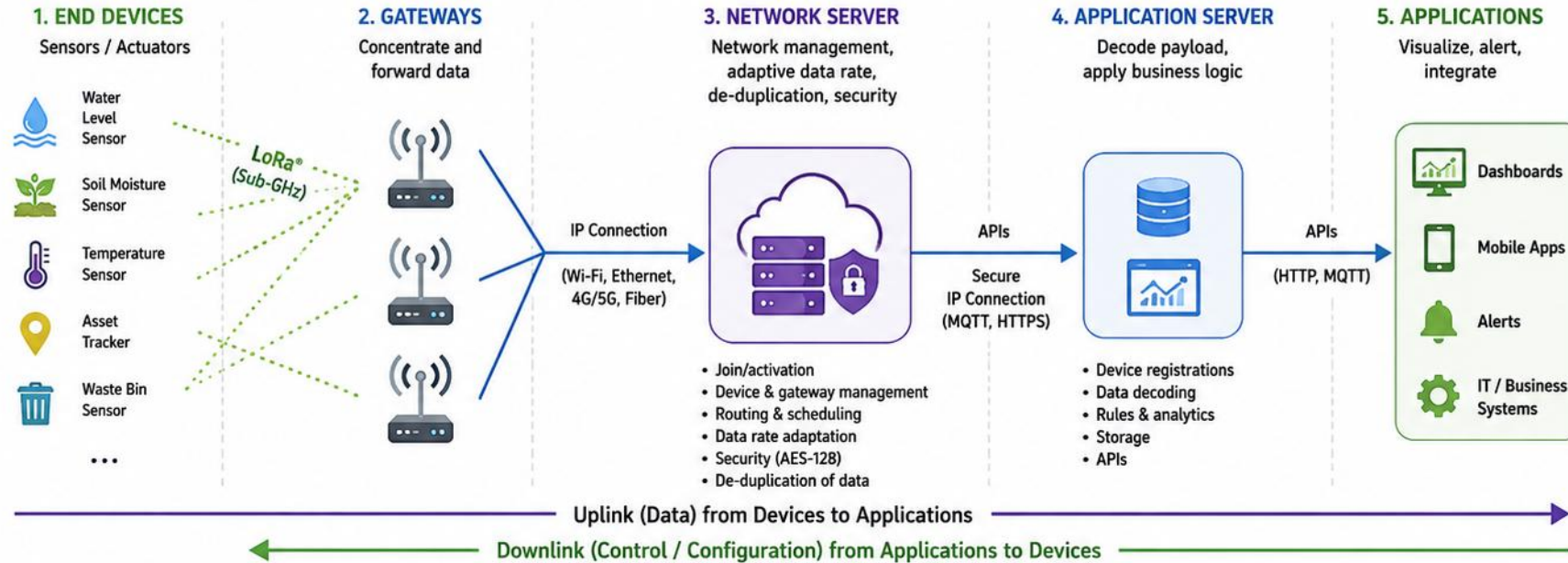
Key Takeaway



If your IoT product needs to work with other brands and platforms, Matter (CHIP) is the standard to build on.

LoRaWAN Architecture and Its Suitability for IoT

LoRaWAN (Long Range Wide Area Network) is an LPWAN technology ideal for low-power, long-range IoT applications.



WHY LoRaWAN IS IDEAL FOR IoT		
	LONG RANGE	2-15+ km in rural areas; hundreds of meters in urban areas. Fewer gateways, lower infrastructure cost.
	LOW POWER	Battery life of 5-10+ years for most use cases. Perfect for battery-powered sensors.
	LOW COST	Low-cost end devices and simple infrastructure. Ideal for large-scale deployments.
	SCALABLE	Supports thousands to millions of devices in a network. Designed for massive IoT.
	SECURE	End-to-end security with AES-128 encryption at network and application layers.
	LICENSE-FREE SPECTRUM	Operates in sub-GHz ISM bands (e.g., 433/868/915 MHz). No subscription fees.
	GOOD PENETRATION	Better penetration through walls, vegetation and underground compared to higher-frequency technologies.



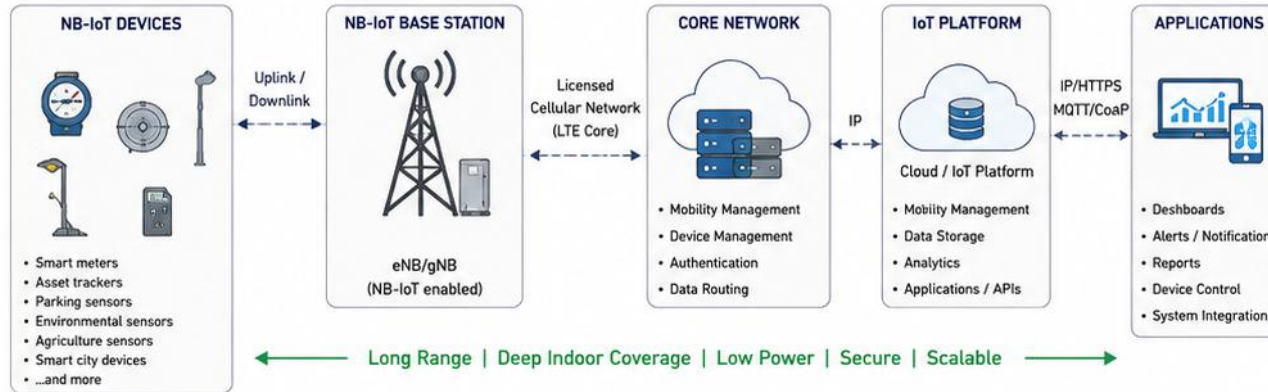
LoRaWAN enables simple, secure, long-range and low-power connectivity – the perfect foundation for massive IoT deployments.

Hiran Ekanayake @ UCSC

NB-IoT (Narrowband IoT)

Low Power, Wide Area Connectivity for Massive IoT

NB-IoT ARCHITECTURE



NB-IoT IN LTE SPECTRUM



HOW NB-IoT WORKS

- Uses a very narrow bandwidth (180 kHz) within existing LTE spectrum.
- Uses OFDMA (downlink) and SC-FDMA (uplink).
- Repeats data transmissions for reliability.
- Designed for small data, infrequent transmissions with very low power.

SUITABILITY FOR IoT

	Long Range Up to 10–15 km in rural areas and several km in urban areas.
	Deep Indoor Coverage Excellent penetration into buildings, basements and remote locations.
	Low Power Consumption Devices can run for 5–10+ years on a single battery.
	Massive Connectivity Supports up to ~100,000 devices per cell.
	High Security Leverages the security of cellular networks (SIM-based, encryption).
	Low Device & Deployment Cost Simple devices and uses existing cellular infrastructure.
	High Availability & Reliability Carrier-grade network with mobility and quality of service.

KEY FEATURES

	Bandwidth	180 kHz
	Peak Data Rate	Uplink: ~62 kbps Downlink: ~26 kbps
	Coverage	Up to 20 dB better link budget than LTE; ideal for deep indoor and remote areas.
	Battery Life	5 – 10+ years (typical use case)
	Latency	Seconds (not for real-time control)
	Device Density	Up to ~100,000 devices per cell
	Mobility	Yes (supports mobility like LTE)
	Standard	3GPP Release 13 and above

NB-IoT vs Other LPWAN Technologies

Feature	NB-IoT	LoRaWAN	LTE-M
Spectrum	Licensed (Cellular)	Unlicensed	Licensed (Cellular)
Range	High (several km)	Very High (up to 15 km+)	High
Indoor Coverage	Excellent	Good	Good
Data Rate	Low (max ~62 kbps)	Very Low (0.3–50 kbps)	Medium (up to 1 Mbps)
Power Consumption	Very Low	Very Low	Low
Mobility Support	Yes	Limited	Yes
Infrastructure	Uses existing cellular network	Requires LoRa gateways	Uses existing cellular network
Use Cases	Mass IoT, metering, tracking, smart cities	Agriculture, rural monitoring	Wearables, tracking, industrial IoT

TYPICAL IoT USE CASES



SUMMARY

NB-IoT is purpose-built for massive IoT deployments that require long range, deep indoor coverage, low power, low cost and secure connectivity. It is ideal for applications with small data packets sent infrequently, such as metering, tracking and smart city sensors.

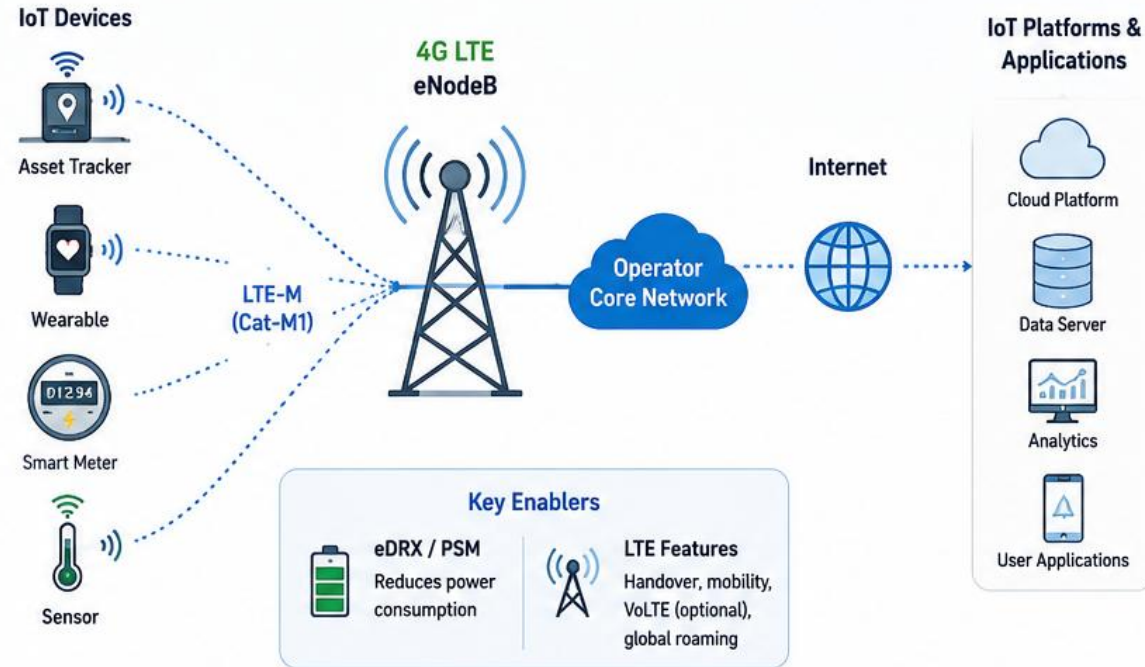
LTE-M (Cat-M1) – Cellular Connectivity Built for IoT

A standards-based 4G (LTE) technology optimized for IoT/M2M applications

KEY FEATURES OF LTE-M (Cat-M1)

- Wide Area Coverage**
Leverages existing 4G LTE infrastructure; excellent indoor and outdoor coverage
- Low Power Consumption**
Extended Discontinuous Reception (eDRX) and Power Saving Mode (PSM) for multi-year battery life
- Balanced Performance**
Higher data rates and lower latency than NB-IoT, supports mobility
- SIM-based Security**
Uses USIM/eSIM with strong authentication and encryption (LTE security)
- Easy Integration**
Based on LTE; works with standard LTE modules and global ecosystems

HOW LTE-M (Cat-M1) WORKS



WHY LTE-M (Cat-M1) IS SUITABLE FOR IoT

- Ubiquitous Coverage**
Uses existing mobile operator networks; works in remote and hard-to-reach areas
- Long Battery Life**
Optimized power saving for devices that transmit small amounts of data infrequently
- Higher Throughput & Lower Latency**
Better user experience for real-time or interactive IoT applications
- Mobility Support**
Supports handover between cells; ideal for moving assets (vehicles, trackers)
- Global Standard**
3GPP LTE standard (Cat-M1) with worldwide ecosystem and roaming
- Secure & Reliable**
Built-in LTE security, network reliability and quality of service

IDEAL IoT USE CASES



LTE-M (Cat-M1) vs. Other Cellular IoT Technologies

Feature	LTE-M (Cat-M1)	NB-IoT (Cat-NB1)	2G/3G	4G LTE
Data Rate (Peak)	Up to ~1 Mbps (DL)	Up to ~250 kbps	Low	High (10s-100s Mbps)
Latency	~50-150 ms	~1-10 s (higher)	High	Low
Power Consumption	Low	Very Low	High	High
Mobility Support	Yes (Handover)	Limited	Yes (2G/3G)	Yes
Voice Support	VoLTE (Optional)	No	Yes (2G/3G)	Yes (VoLTE)
Best For	Mobile / Data-focused IoT	Ultra low power static IoT	Legacy / Voice	High bandwidth applications



LTE-M (Cat-M1) delivers the right balance of coverage, power efficiency, performance, mobility and cost, making it one of the most versatile and widely adopted cellular IoT technologies.

View Full Presentation @ IOT5G



5G AND ITS SUITABILITY FOR IoT

WHAT IS 5G?

5G (Fifth Generation) is the next generation mobile network technology that delivers high data rates, ultra-low latency, massive capacity and high reliability. It provides an ideal wireless foundation for a wide range of IoT applications.



KEY CAPABILITIES OF 5G



High Data Rate

Up to 10 Gbps peak data rate
(> 100x faster than 4G)



Ultra-Low Latency

As low as 1 ms air interface latency
(~10x lower than 4G)



Massive Connectivity

Up to 1 million devices per km²
(>10x more than 4G)



High Reliability & Availability

99.999% reliability for mission critical services



Energy Efficiency

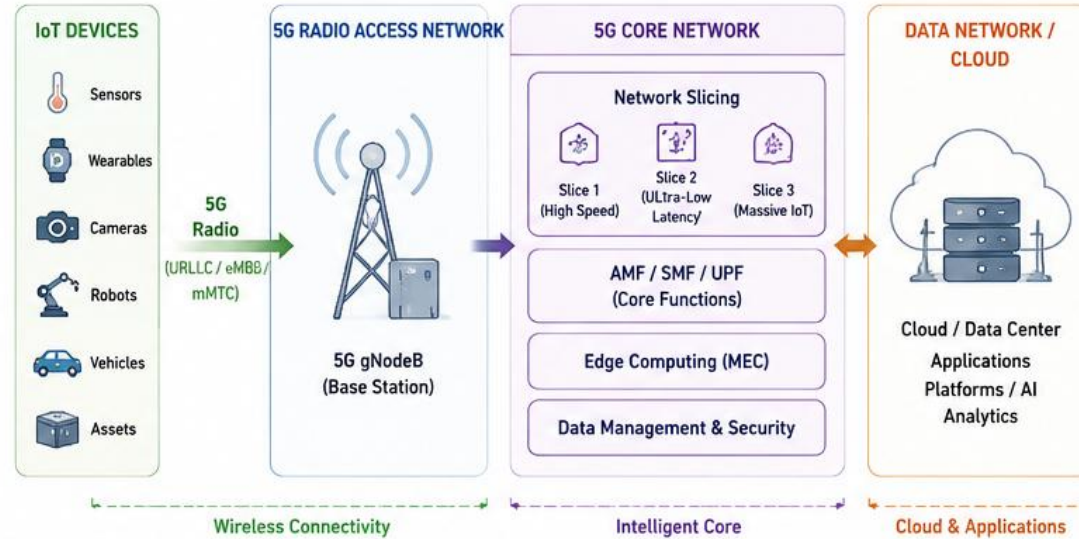
Improved battery life for IoT devices



Network Slicing

Logical separation of networks for different services and QoS

5G IoT NETWORK ARCHITECTURE



5G SERVICE CATEGORIES FOR IoT

eMBB (Enhanced Mobile Broadband)



High data rate for video streaming, smart cameras, AR/VR, drones

URLLC (Ultra Reliable Low Latency Communication)



Ultra low latency and high reliability for industrial automation, robotics, remote surgery, autonomous systems

mMTC (Massive Machine Type Communication)



Massive number of low-power devices for smart meters, tracking, sensors

5G ENABLED IoT USE CASES



Smart Manufacturing

- Real-time machine control
- Predictive maintenance
- AGV/AMR connectivity



Autonomous Vehicles

- Ultra-low latency control
- Vehicle-to-everything (V2X)
- High reliability



Smart Cities

- Smart traffic management
- Surveillance & safety
- Smart lighting & waste mgmt.



Smart Healthcare

- Remote surgery
- Connected ambulances
- Wearables monitoring



Utilities & Energy

- Smart grid automation
- Fault detection
- Remote asset monitoring



Agriculture

- Precision farming
- Drone monitoring
- Livestock tracking



Logistics & Tracking

- Asset & container tracking
- Cold chain monitoring
- Fleet management

5G VS OTHER TECHNOLOGIES FOR IoT

Technology	Typical Range	Data Rate (Peak)	Latency	Device Density	Power Consumption	Best Suited IoT Applications
5G (mMTC / URLLC / eMBB)	Up to several km	Up to 10 Gbps	~1 ms	Up to 1M devices/km ²	Medium (with optimizations)	Industrial IoT, smart cities, autonomous systems, video IoT, massive deployments
4G LTE	Up to several km	Up to 1 Gbps	20–30 ms	~100K devices/km ²	Medium	General mobile IoT, tracking, telematics
NB-IoT	Up to 10+ km	Up to 250 kbps	1.6–10 s	~100K devices/km ²	Very Low	Smart meters, simple sensors, long battery life
LoRaWAN	Up to 15+ km	0.3–50 kbps	Seconds	~10K devices/km ²	Very Low	Agriculture, tracking, asset monitoring
Wi-Fi 6/6E	30–100 m	Up to 9.6 Gbps	~10–20 ms	~1K devices/AP	High	High data rate IoT, cameras, gateways
Bluetooth LE	10–100 m	Up to 2 Mbps	~6–30 ms	~connections limited	Very Low	Wearables, beacons, short range sensors

WHY 5G IS WELL SUITED FOR IoT

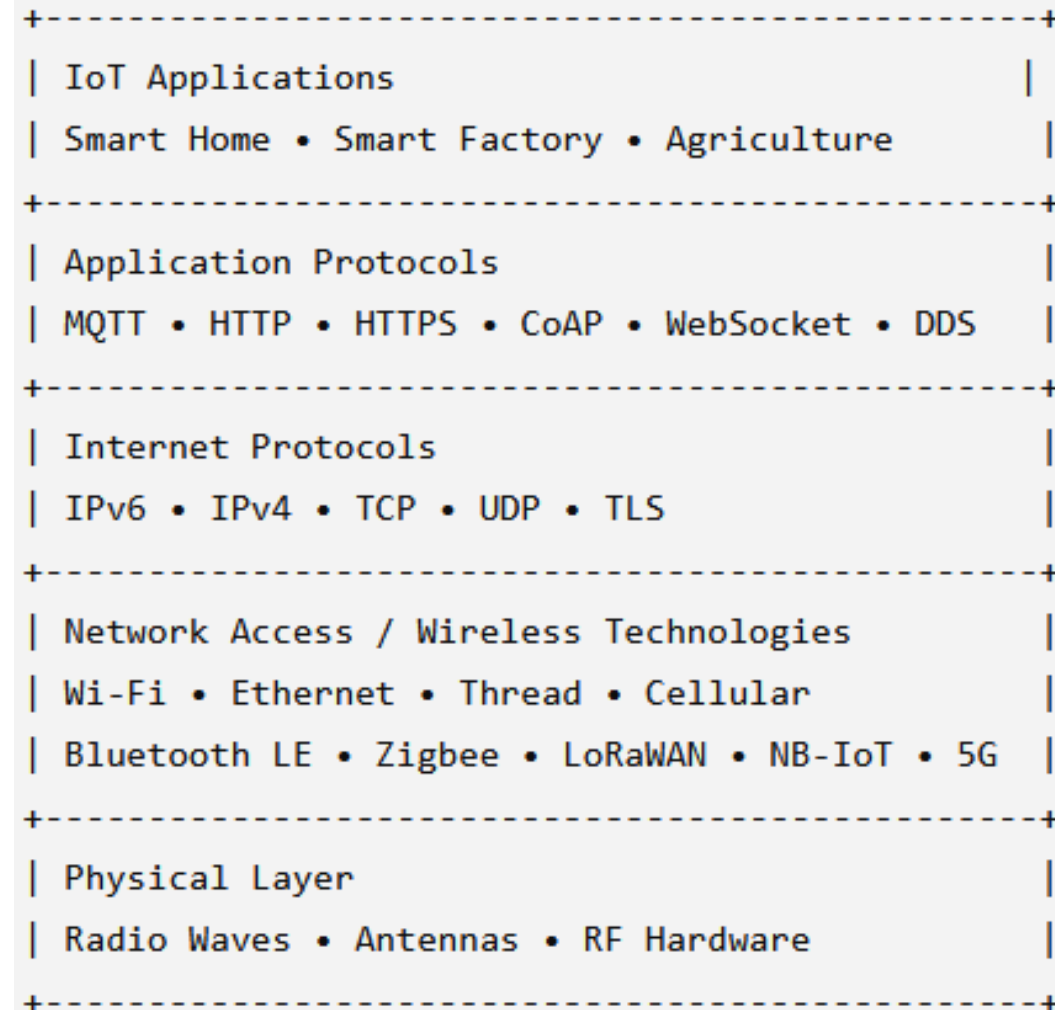
- ✓ Supports diverse IoT requirements through eMBB, URLLC, mMTC.
- ✓ Handles massive number of devices with efficient spectrum use.
- ✓ Enables real-time, mission-critical and high-bandwidth applications.
- ✓ Network slicing ensures dedicated performance and security.
- ✓ Edge computing with 5G reduces latency and bandwidth usage.
- ✓ Future-proof platform for AI-driven and autonomous IoT systems.

Summary

- **Wi-Fi** provides high-speed connectivity for data-intensive applications.
- **Bluetooth Low Energy** enables low-power personal and wearable devices.
- **Zigbee** and **Thread** support reliable, low-power mesh networks.
- **Matter (formerly CHIP)** provides interoperability across smart home ecosystems by operating over IP-based transports such as Thread, Wi-Fi, and Ethernet.
- **LoRaWAN**, **NB-IoT**, and **LTE-M** connect low-power devices over long distances.
- **5G** enables high-capacity, low-latency applications such as smart factories and autonomous vehicles.
- **NFC** supports secure, short-range interactions such as payments and device commissioning.

IoT Communication Stack

- IoT communication technologies can be organized into several layers, similar to the Internet protocol stack.
- Different Technologies Have Different Roles:
 - E.g., sending an email: Wi-Fi is the **road**. IP is the **postal system**. MQTT is the **language used to exchange messages**.
- The **three distinct categories of communication technologies** in IoT:
 - **Wi-Fi, Bluetooth, LoRaWAN, and 5G** answer "How do the bits travel?"
 - **TCP, UDP, and IP** answer "How are the bits transported across networks?"
 - **MQTT, HTTP, CoAP, and Matter** answer "How do applications communicate and understand each other's messages?"

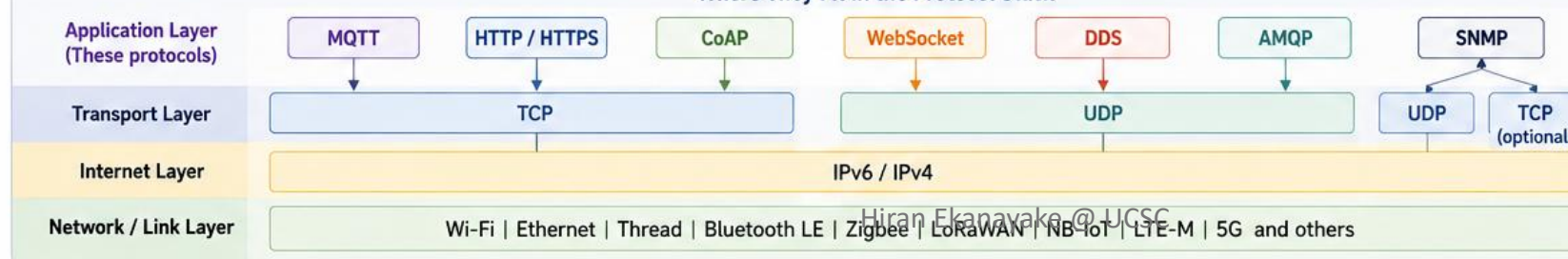


Differences Between Common Internet Protocols (IoT Focus)

Application layer protocols that run over TCP or UDP (except where noted).

Feature	MQTT	HTTP / HTTPS	CoAP	WebSocket	DDS	AMQP	SNMP
Full Name							
Transport	MQTT (Message Queuing Telemetry Transport)	HyperText Transfer Protocol / Secure	Constrained Application Protocol	WebSocket Protocol	Data Distribution Service	Advanced Message Queuing Protocol	Simple Network Management Protocol
Model	TCP (default 1883) TLS (8883)	TCP 80 (HTTP) / 443 (HTTPS)	UDP (5683) DTLS (5684)	TCP (80 / 443)	UDP (default) or TCP	TCP (5672) TLS (5671)	UDP (161, 162) TCP (rare)
Overhead / Header Size	Publish / Subscribe via Broker	Request / Response	Request / Response (REST-like)	Full Duplex (Bi-directional)	Publish / Subscribe (Data-Centric)	Queue / Message Oriented	Manager / Agent (Request / Response)
Connection	Very Low (2 bytes fixed header)	High (Text based)	Very Low (4 bytes base header)	Low-Medium (2-14 bytes)	Medium-High (20+ bytes)	High (Complex headers)	Low-Medium (ASN.1 BER encoded)
Connection	Persistent TCP Keep-Alive	Short-lived (can use keep-alive)	Connectionless (UDP) or DTLS	Persistent TCP (Upgraded)	Persistent (Peer discovery)	Persistent TCP (Channels)	Short-lived (Usually)
Reliability	High (TCP) QoS 0,1,2 levels	High (TCP)	Best Effort (UDP) Confirmable option	High (TCP)	High + QoS Configurable	High (TCP) Transactions	Low-Medium (UDP default)
Best For	IoT telemetry, low bandwidth, many devices	Web services, configuration, interoperability	Constrained devices, low power, low bandwidth	Real-time bi-directional communication	Real-time systems, industrial, robotics, distributed control	Enterprise messaging, reliable queuing	Network management, monitoring
Typical Use Cases	 Sensors, gateways, telemetry, smart home	 REST APIs, device config, web servers	 Smart meters, wearables, LPWAN	 Dashboards, chat, live updates	 Robotics, autonomous systems, aviation	 Message brokers, backend systems	 Routers, switches, network devices
Pros	Lightweight, scalable, low bandwidth, simple	Universal support, easy to use, firewall friendly	Very lightweight, ideal for constrained nodes	Real-time, full duplex over a single connection	High performance, scalable, rich QoS, real-time	Reliable messaging, queuing, routing, transactions	Simple, widely supported for network devices
Cons	Requires broker, not for large file transfers	High overhead, not efficient for frequent small data	UDP unreliable, security (DTLS) costly	More complex than HTTP, not cacheable	Complex, larger footprint, harder to implement	Heavy protocol, more overhead, higher complexity	Not secure (v1/v2), limited functionality

Where They Fit in the Protocol Stack



Choose the protocol based on application requirements: bandwidth, latency, power, reliability and network scale.



Use TLS/DTLS for security where possible.



MQTT, CoAP and other lightweight protocols are optimized for resource constrained IoT devices.

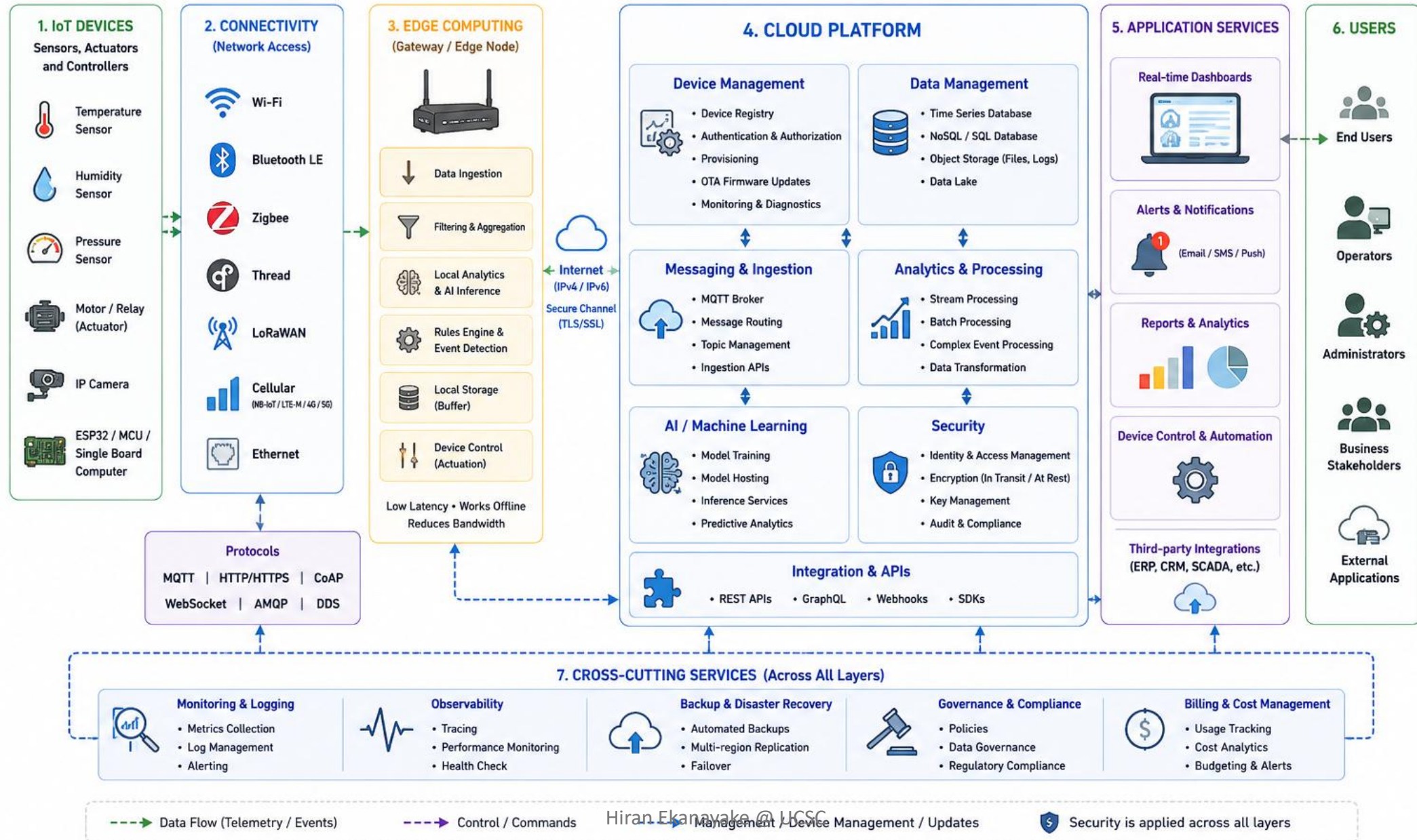
IoT Cloud Computing and Cloud Architecture

What is IoT Cloud Computing?

- **IoT Cloud Computing** is the use of cloud infrastructure to collect, store, process, analyze, and manage data generated by IoT devices while providing applications and services over the Internet.
- Instead of every device performing complex computations, the cloud provides virtually unlimited computing resources.
 - E.g., an ESP32 cannot efficiently store years of sensor data or train AI models. The cloud overcomes these limitations.
- Cloud platforms provide: Massive storage, High-performance computing, Artificial intelligence services, Databases, Global accessibility, Device management

Components of IoT Cloud Architecture

End-to-end view showing devices, connectivity, edge computing and cloud platform



Components of the Cloud Architecture

- **1. IoT Devices**

- These are the physical devices that generate data.
- E.g., ESP32, Raspberry Pi, Smart meters, Wearable devices, Industrial sensors
- Functions: Data collection, Local control, Sensor interfacing

- **2. Connectivity**

- Devices communicate using technologies such as Wi-Fi, Ethernet, Bluetooth Gateway, LoRaWAN Gateway, NB-IoT, LTE-M, 5G
- Application protocols include MQTT, HTTP, CoAP

- **3. Edge Computing Layer**

- Edge computing processes data close to where it is generated to reduce bandwidth and cloud workload.
- Examples of edge processing: Noise filtering, Data compression, AI inference, Event detection, Local automation
- E.g., Instead of transmitting every sensor reading, 25.1, 25.2, 25.3, 25.2, 25.1, etc., the edge device may send Average = 25.2°C or ALERT: Water Level < 20%.

Components of the Cloud Architecture Cont.

- **4. Cloud Platform**

- The cloud platform is the core of the IoT ecosystem.
- Typical responsibilities include:
 - **Device Management** (Register devices, Authentication, Firmware updates (OTA), Monitoring device health),
 - **Data Storage** (Cloud databases store Sensor readings, Images, Events, Logs),
 - **Data Analytics** (Cloud analytics can Detect trends, Predict failures, Generate reports, Calculate statistics)
 - **Artificial Intelligence** (Cloud AI is used for Model training, Image recognition, Predictive maintenance, Forecasting, Large-scale analytics)
 - **Security** (Security services include Authentication, Authorization, Encryption, Identity management, Secure communication)

IoT Cloud Platforms

- An **IoT Cloud Platform** is a software platform that enables IoT devices to connect, communicate, store data, perform analytics, and provide applications for monitoring and control.
- It acts as the **Platform Layer** in an IoT architecture, sitting between connected devices and end users.
- Most IoT cloud platforms provide: Device registration | Authentication | MQTT broker | Data storage | Dashboards | Data visualization | Analytics | Alerts and notifications | Remote device control | OTA firmware updates | APIs | User management

Dashboard

Devices

Data Explorer

Alerts

Rules & Automations

Locations

Users

Reports

Integrations

Settings

Quick Actions

Add Device

Create Rule

Send Notification

Export Data

May 18 – May 24, 2025



+ Add Widget



Average Water Level

68%

↑ 12% from last week



Total Tanks

12

Online: 11 Offline: 1



Alerts

3

[View all alerts →](#)



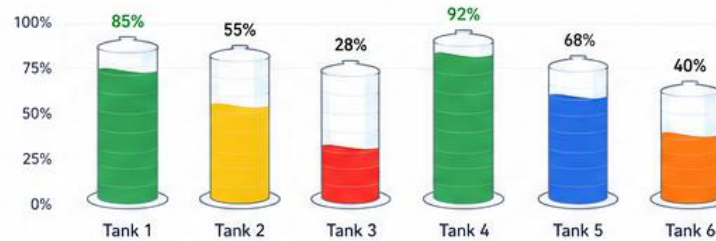
Total Consumption (This Week)

18,540 L

↑ 8.5% from last week

Water Level Overview

All Tanks



Location Map



Recent Alerts

[View all](#)



Tank 3 – Low Water Level
Water level is below 30%
Today, 09:15 AM



Tank 6 – High Usage
Abnormal usage detected
Today, 08:40 AM



Tank 2 – Sensor Offline
No data received for 15 min
Today, 07:55 AM

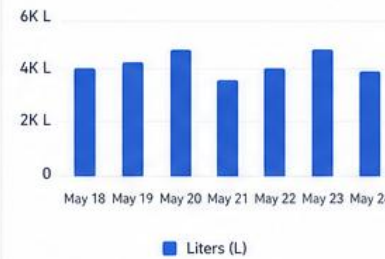
Water Level Trend

Tank 1



Daily Consumption

All Tanks



Tank 1 – Real Time Data



My Devices

[View all](#)

Tank 1	Online	85%	
Tank 2	Online	55%	
Tank 3	Online	28%	
Tank 4	Online	92%	
Tank 5	Online	68%	
Tank 6	Online	40%	

1 Offline Device

System Activity

Time	Device	Event	Status
10:30:45 AM	Tank 1	Telemetry Data Received	Success
10:30:21 AM	Tank 6	Telemetry Data Received	Success
10:29:58 AM	Tank 2	Telemetry Data Received	Success
10:29:30 AM	Tank 3	Sensor Offline	Warning
10:29:10 AM	Tank 4	Firmware Update	Success

Data Flow



Hiran Ekanayake @ UCSC

1. Commercial Enterprise Platforms

- Designed for large-scale industrial and enterprise deployments.
- E.g., Amazon Web Services IoT, Microsoft Azure IoT, Google Cloud IoT services, IBM Maximo Application Suite, Oracle IoT solutions
- Advantages: Extremely scalable, Enterprise-grade security, Global infrastructure, AI and machine learning integration, Professional support
- Disadvantages: Complex to learn, Can become expensive, Requires cloud knowledge
- Best for: Smart factories, Utilities, Large enterprises, Research projects with many devices

2. IoT Platform as a Service (PaaS)

- These platforms simplify IoT application development by providing managed services.
- E.g., Ubidots, Datacake, Blynk, Losant, Arduino Cloud
- Advantages: Easy setup, Built-in dashboards, Mobile applications, Little server management
- Disadvantages: Monthly subscription for advanced features, Less customization
- Best for: Universities, Prototypes, Small businesses, Rapid development

3. Open-Source IoT Platforms

- These platforms can be installed on your own computer, virtual machine, or cloud server.
- E.g., ThingsBoard, Node-RED, Eclipse Foundation Mosquitto (MQTT broker), EMQ Technologies EMQX Community Edition, InfluxData InfluxDB, Grafana Labs Grafana, OpenRemote
- Advantages: Free, Fully customizable, No vendor lock-in, Suitable for teaching
- Disadvantages: Requires Linux/server knowledge, Maintenance is your responsibility
- Best for: Universities, Research laboratories, Developers, Hobbyists wanting full control

4. DIY / Maker Platforms

- These are optimized for makers, students, and hobbyists.
- Popular platforms: Blynk, Arduino Cloud, Adafruit IO, ThingSpeak, Datacake
- Advantages: Very easy to use, Mobile apps, Excellent documentation, ESP32 support, Free tiers
- Disadvantages: Limited scalability, Storage limits, Device limits

5. Industrial IoT Platforms

- Focused on Industry 4.0 applications.
- Examples: Siemens Industrial Edge, PTC ThingWorx, AVEVA CONNECT, Schneider Electric EcoStruxure
- These integrate with: PLCs, SCADA, ERP, Manufacturing Execution Systems (MES)

Comparison

Platform Type	Cost	Ease of Use	Scalability	Best For
Enterprise Cloud	High	Medium	Excellent	Industry
PaaS	Medium	High	Good	SMEs, universities
Open Source	Free	Medium	Excellent	Developers, research
Maker Platforms	Free/Low	Very High	Moderate	Students, hobbyists
Industrial Platforms	High	Medium	Excellent	Manufacturing

Recommended Platforms for Different Users

User	Recommended Platforms
Beginners	Blynk, Arduino Cloud
Hobbyists	Blynk, Adafruit IO, ThingSpeak
DIY Makers	Blynk, Adafruit IO, Node-RED + Mosquitto
University Students	ThingSpeak, Blynk, Arduino Cloud, ThingsBoard Community
Research Projects	ThingsBoard, Node-RED, Grafana, InfluxDB
Universities	ThingsBoard Community + Mosquitto + Grafana
Startups	Datacake, ThingsBoard Cloud, AWS IoT
Large Enterprises	AWS IoT, Azure IoT, Google Cloud

The Blynk IoT Platform

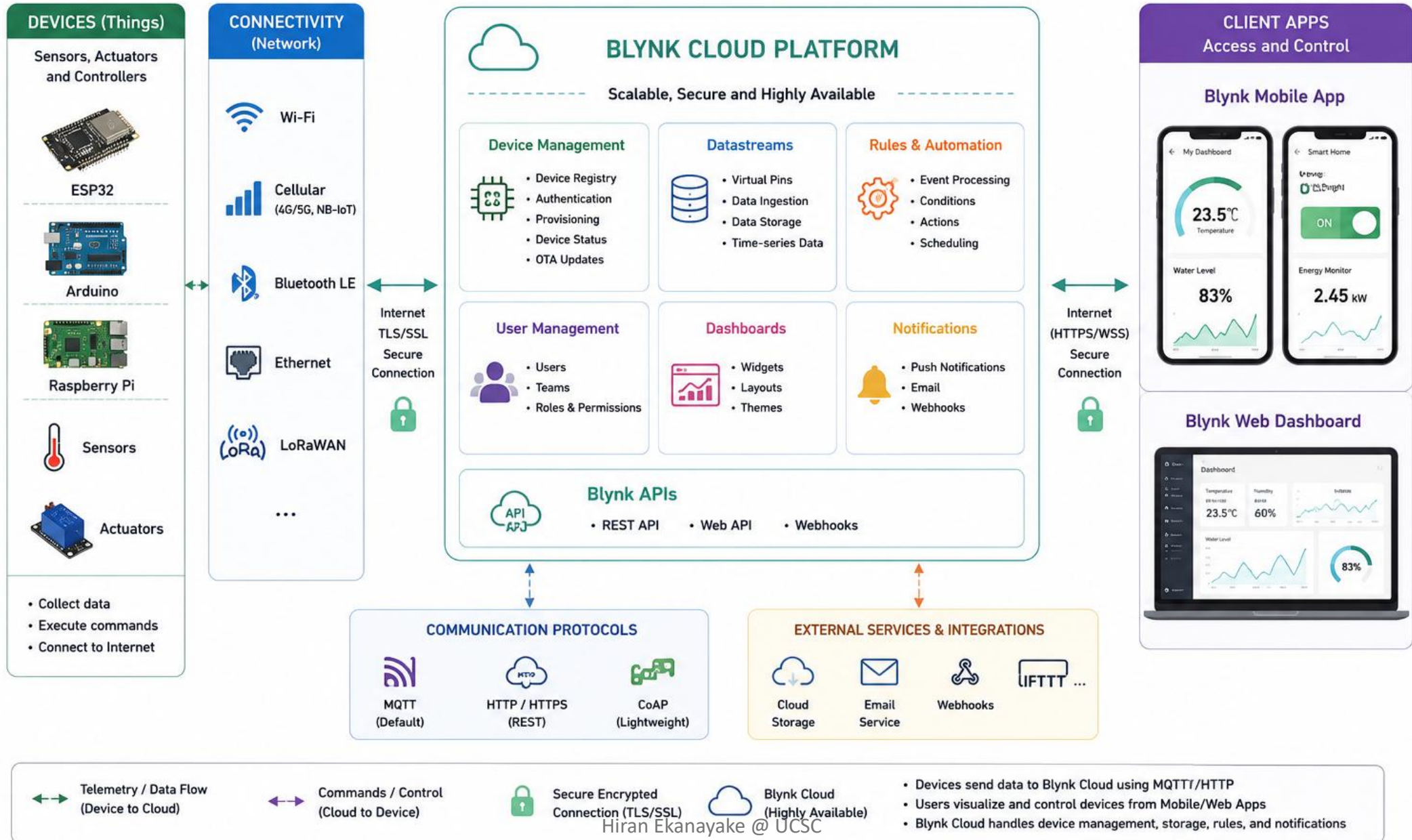
What is Blynk?

<https://www.blynk.io/>

- **Blynk** is a cloud-based Internet of Things (IoT) platform that allows users to **build, monitor, and control IoT devices remotely** using mobile applications and web dashboards without having to develop their own cloud infrastructure.
- It is one of the most popular IoT platforms for: Students, Hobbyists, DIY makers, Universities, Rapid prototyping, Small commercial IoT products
- The platform supports popular development boards such as: ESP32, ESP8266, Arduino Uno R4 WiFi, Raspberry Pi 5 (via Linux applications)
- Watch: <https://www.youtube.com/watch?v=-bZp36zWQ1Y>

BLYNK ARCHITECTURE

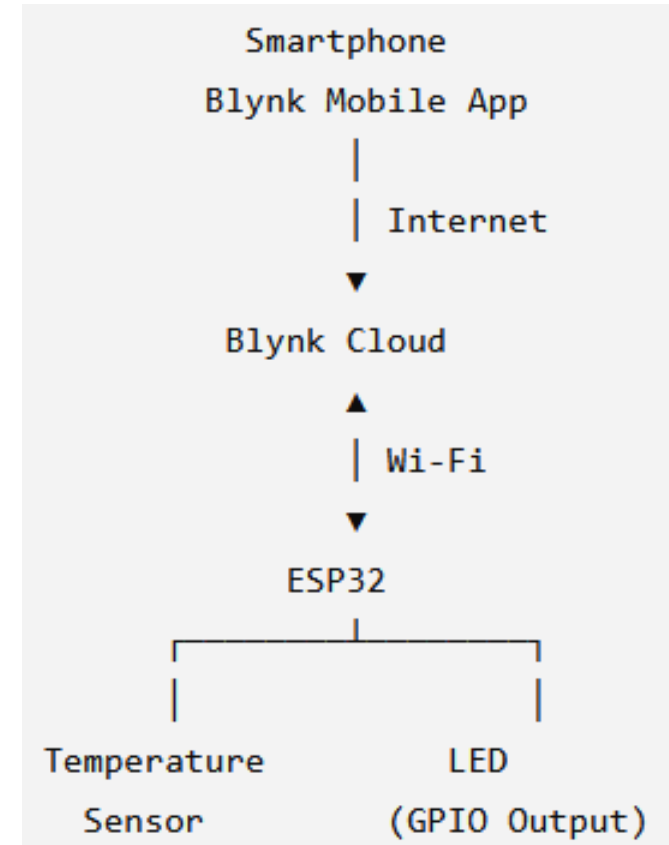
Connect – Manage – Visualize – Automate



Developing a Simple ESP32- Based IoT Application Using Blynk

The Application

- One of the easiest ways to learn IoT is to build a simple application that reads data from an ESP32 and displays it on a smartphone or web dashboard using **Blynk**.
- In this example, we will create an application that:
 - Reads the ESP32's built-in temperature (or a simulated sensor value).
 - Sends the value to the Blynk Cloud.
 - Displays the value on a mobile app and web dashboard.
 - Optionally controls an LED from the app.



Steps

- Step 1 – Create a Blynk Account
 - Create a free account on the Blynk platform.
 - Log in to the Blynk Console.
 - Verify your email address.

- Get Started
- Dashboards
- Custom Data
- Developer Zone
- Devices
- Automations
- Users
- Organizations
- Locations
- Snapshots
- Fleet Management
- In-App Messaging
- Predictive Maintenance

Get Started With Blynk

YOUR PLAN DETAILS

0/5

Devices

0/100000

Device messages

0/10

Templates created

1/1

Team members

SUGGESTED FOR YOU



Create your app in minutes with Blueprints

Use a pre-built template with all the essentials to get your device up and running instantly.

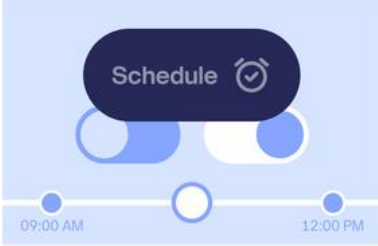
[View Blueprints](#)



Customize your app's look and feel

Skip the time-consuming app review process and customize your app with no coding required.

[Learn More](#)



No-code automations

Automate device actions, alerts, and conditions without writing extra code—across your IoT ecosystem.

[Try Now](#)



Events, alerts and notifications

Track important events, receive real-time alerts, and configure notifications for your devices.

[Learn More](#)

Activate your devices in Blynk

Connect via Blynk.Edgent, Blynk.NCP, MQTT, HTTPs API, LoRaWAN, the Blynk Library, or import from third-party cloud services.

[Connect A Device](#) [View Compatible Devices](#)

Need help? Ask the AI assistant

Blynk's AI assistant can help you set up devices, configure datastreams, build dashboards, and troubleshoot issues. Click the AI button in the top right corner to get started.

[Ask AI](#)

Welcome, Hiran

Here's a quick checklist to get you up to speed with Blynk Platform:

- ☒ [Intro to Blynk](#)
- ☐ [Add template or blueprint](#)
- ☐ [Connect first device](#)
- ☐ [Build a dashboard](#)
- ☐ [Download Blynk mobile app](#)

Blynk Onboarding #2: Con & Blueprints

Learn how to create a custom device template or use a ready-made blueprint

[Use a blueprint](#) [Create template](#)

Steps Cont.

- Step 2 – Create a Template
 - E.g., Template Name: ESP32 Demo, Hardware: ESP32, Connection: Wi-Fi
 - The template defines the data model for all devices using it.

Create New Template

NAME: ESP32 Demo (10 / 50)

HARDWARE: ESP32

CONNECTION TYPE: WiFi

DESCRIPTION: Description (0 / 128)

Buttons: Cancel, Done

ESP32 DEMO

Home

Datastreams

Dashboard

Events & Notifications

Connection Lifecycle

Data Converters

Metadata

Assets

App Dashboard

User Guides

Automation Templates

Voice Assistants

Steps Cont.

- Step 3 – Create Datastreams
 - Create two Virtual Datastreams.

Datastream	Type	Purpose
V0	Double	Temperature
V1	Integer	LED Control

Datastreams

All 2 Integer Double String Enum Location Digital Analog

☐	Pin	Name	Data Ty...	Semantics	Min	Max	Default	Units	Use in Autom...
	V0	Double V0	Double	Temperature [Celsius]	-40	125	0	°C	None
	V1	Integer V1	Integer	LED	0	1	0	-	None

New Datastream

General

Use in Automations

Advanced Settings

Code

NAME

Double V0

9 / 128

PIN

V0

DISPLAY NAME

Double V0

SEMANTIC TAGS (OPTIONAL)

Temperature [Celsius] × +

Help us recognize the meaning of your data (e.g., temperature, battery, RSSI) to power smarter dashboards ar

DATA TYPE

Double

UNITS

Celsius, °C

MIN

-40

MAX

125

DECIMALS

#.##

DEFAULT VALUE

0

Steps Cont.

- Step 4 – Create a Device
 - From the template: New Device ➔ From Template
 - Blynk generates: Device Name & Authentication Token
 - Example: Auth Token, abc123xyz....
 - The token uniquely identifies the ESP32.

ESP32 DEMO

 Home

 Datastreams

 Dashboard

 Events & Notifications

 Connection Lifecycle

 Data Converters

 Metadata



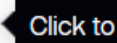
Home

What's next?

☐ [Configure template](#)

☐ [Set Up Datastreams](#)

☐ [Set up the Web Dashboard](#)

☐ [Add first Device](#) 

Steps Cont.

- Step 5 – Design the Dashboard
 - Add widgets to display temperature and turn the LED ON/OFF
 - Gauge → Temperature (V0)
 - Switch → LED Control (V1)
- Step 6 – Install Arduino Libraries
 - In the Arduino IDE, install:
 - Blynk
 - ESP32 Board Package (if not already installed)

Dashboard

 This is how the device page will look like for actual devices.



Device Name ● Online

[Device Owner](#) [Company Name](#)



1h

6h

1d

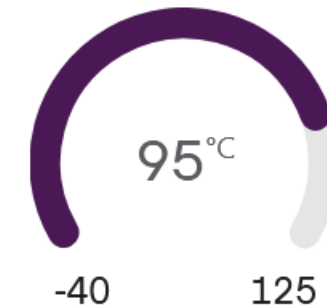
1w

1mo ●

3mo ●



Double V0 (V0)



Integer V1 (V1)



Steps Cont.

- Step 7 – Configure the Program
- Step 8 – Initialize Wi-Fi
 - Once connected, the device automatically appears online in the Blynk Console.
- Step 9 – Read Sensor Data
 - The ESP32 periodically reads a sensor.
- Step 10 – Send Data to Blynk
 - The value is written to Virtual Pin V0 and the gauge immediately updates.
- Step 11 – Receive Commands
 - Suppose the user presses a switch. The ESP32 receives the command and changes the GPIO output.

```
#define BLYNK_TEMPLATE_ID "YOUR_TEMPLATE_ID"
#define BLYNK_TEMPLATE_NAME "ESP32 Demo"
#define BLYNK_AUTH_TOKEN "YOUR_AUTH_TOKEN"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

//-----
// WiFi Credentials
//-----

char ssid[] = "YOUR_WIFI_NAME";
char pass[] = "YOUR_WIFI_PASSWORD";
```

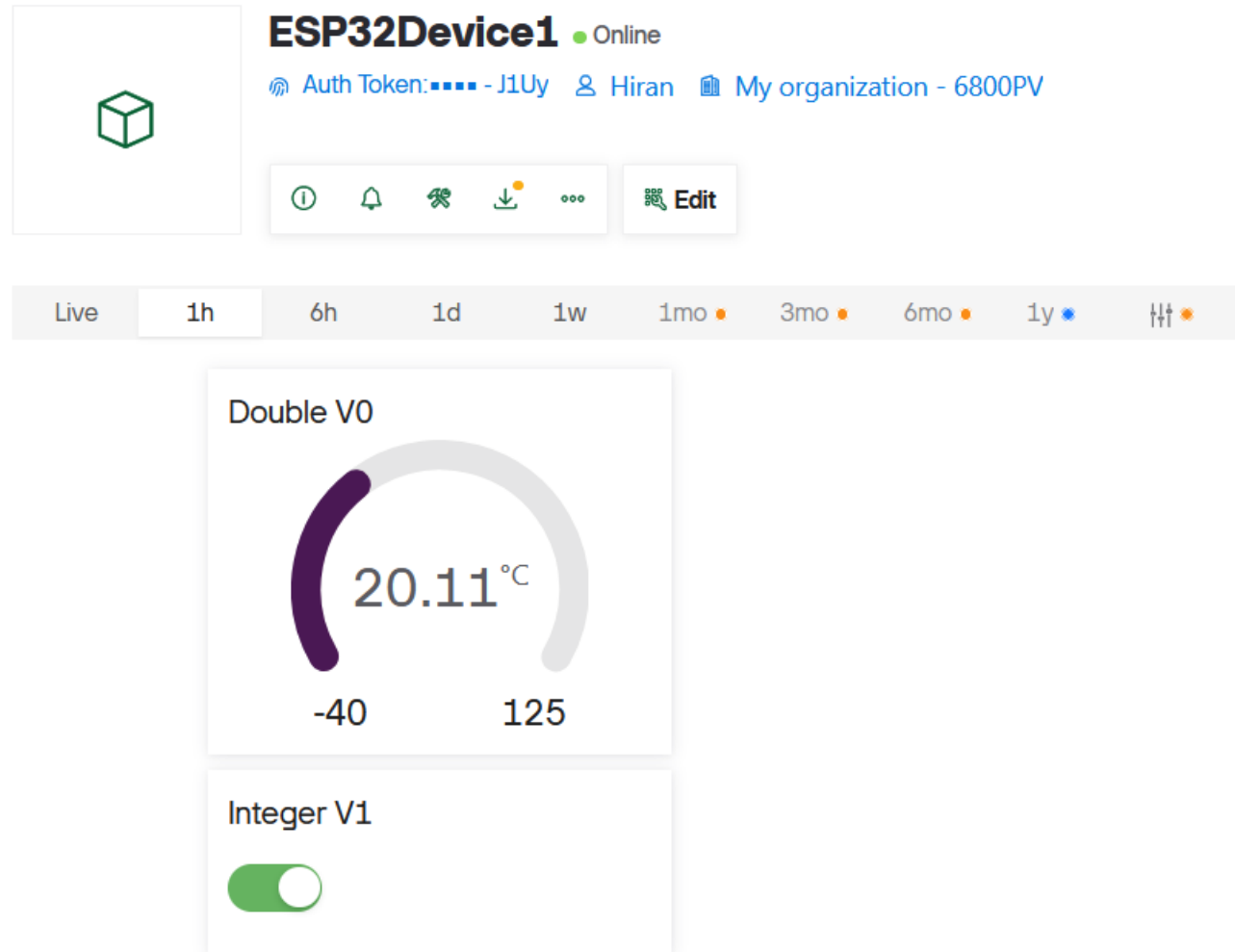
The Program

```
1  #define BLYNK_TEMPLATE_ID "YOUR_TEMPLATE_ID"
2  #define BLYNK_TEMPLATE_NAME "ESP32 Demo"
3  #define BLYNK_AUTH_TOKEN "YOUR_AUTH_TOKEN"
4
5  #include <WiFi.h>
6  #include <BlynkSimpleEsp32.h>
7
8  //-----
9  // WiFi Credentials
10 //-----
11 char ssid[] = "YOUR_WIFI_NAME";
12 char pass[] = "YOUR_WIFI_PASSWORD";
13
14 //-----
15 // Built-in LED
16 //-----
17 const int LED_PIN = 2;
18
19 //-----
20 // Blynk Timer
21 //-----
22 BlynkTimer timer;
23
24 //-----
25 // Simulated Temperature
26 //-----
27 float temperature = 25.0;
28
```

```
29 //-----
30 // Send data to Blynk every second
31 //-----
32 void sendSensor() {
33     // Simulated changing temperature
34     temperature = 25.0 + 5.0 * sin(millis() / 5000.0);
35
36     Serial.print("Temperature: ");
37     Serial.println(temperature);
38
39     // Send to Virtual Pin V0
40     Blynk.virtualWrite(V0, temperature);
41 }
42
43 //-----
44 // Receive Switch Value from App
45 //-----
46 BLYNK_WRITE(V1) {
47     int state = param.asInt();
48
49     digitalWrite(LED_PIN, state);
50
51     Serial.print("LED State: ");
52     Serial.println(state);
53 }
54
```

```
55 //-----
56 // Setup
57 //-----
58 void setup() {
59     Serial.begin(115200);
60
61     pinMode(LED_PIN, OUTPUT);
62     digitalWrite(LED_PIN, LOW);
63
64     Serial.println();
65     Serial.println("Connecting to WiFi...");
66
67     WiFi.begin(ssid, pass);
68
69     while (WiFi.status() != WL_CONNECTED) {
70         delay(500);
71         Serial.print(".");
72     }
73
74     Serial.println("\nWiFi Connected!");
75
76     Blynk.config(BLYNK_AUTH_TOKEN);
77     Blynk.connect();
78
79     timer.setInterval(1000L, sendSensor);
80
81     Serial.println("Connected!");
82 }
83
84 //-----
85 // Main Loop
86 //-----
87 void loop() {
88     Blynk.run();
89     timer.run();
90 }
```


The Dashboard



Compare and choose what works best for your project

	Free	Starter	Prototype	Production
	FOR EXPLORATION \$0/month	FOR HOBBY AND EARLY PILOTS \$29/month	FOR PROTOTYPING AND SMALL FLEETS \$99/month	FOR SCALING DEPLOYMENTS \$199/month
				100 ▾
MONTHLY ☒ YEARLY ☐ Save 20%	5 Devices and 1 User	10 Devices and 1 User	50 Devices and Users	100 Devices and Users
Devices included in the plan Device is a microcontroller with unique Auth Token (for example single ESP32)	5	10	50	100-1000
Users included in plan	1	1	50	100-1000
Messages monthly	100,000	10,000,000	Unlimited	Unlimited
Historical data storage	1 week	1 month	6 months	12+ months of historical data storage
Dashboards Real-time overview of multiple devices, including different device types, with data aggregation, historical trends, and control	1 dashboard per organization	1 dashboard per organization	5 dashboards per organization	5 dashboards per organization
Automations per organization Automation is a scenario where the device automatically performs one or more actions based on a condition	5	10	50	100
Data converters		✓	✓	✓
Create and manage sub-organizations		✓	✓	✓